

The Roots of Faith: Early Life Droughts and Religious Commitment in Africa

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Abstract

Can adverse environmental shocks durably shape religious life? Combining individual-level data from three Afrobarometer rounds across 37 African countries with over a century of high-resolution drought data, I exploit variation in drought exposure across birth cohorts within localities and find that individuals exposed to stronger droughts are more likely to be members of a religious group, while their religious affiliation is unchanged. While a short-run effect is present for adults of all ages, the persistent, long-term impact is driven specifically by exposure during the formative teenage years. This effect is concentrated among individuals who demonstrably grew up where they are observed — those living in their ethnic homeland or among their own ethnic group — and teenage drought does not predict who stays, ruling out migratory sorting. The estimate is robust to locality-specific cohort trends, which also remove the one placebo failure in the design, and to randomization inference that reassigns entire drought histories across localities. The response is specific to religious institutions: secular community groups respond only at the threshold of adulthood and more weakly, and trust in state and traditional authorities does not move. Consistent with religious groups providing informal insurance, the effect is concentrated in poorer countries and is equally strong where droughts are rare and where they are common, suggesting little adaptation. The results identify a demand-side channel through which climate volatility builds durable religious institutions.

Keywords: Religion, Development, Life-cycle, Shocks.

JEL Classification: Z10, Z12, Q54, O13

1. Introduction

Among its multifaceted roles, religion influences cultural norms and values (Becker et al., 2021) and shapes societal decision-making (Nunn, 2022). As social and cultural behavior impacts economic activity, attention to religion is crucial in examining the dynamics that determine economic development. This consideration is particularly relevant for developing countries, where — in general — religion is much more prevalent (Pew Research Center, 2018). Consequently, there is extensive evidence documenting a causal negative relationship between different facets of religion and economic development (Andersen and Bentzen, 2022; McCleary and Barro, 2006; Campante and Yanagizawa-Drott, 2015; Montero and Yang, 2022; Lawson, 2018). As religious presence, adherence, and religiosity vary significantly across nations, individuals, and historical periods (Pew Research Center, 2018), understanding the drivers behind this variation in religiosity and religious practices can provide key insight into the determinants of economic development.

Various factors have been explored in understanding individuals' inclination towards religion or religious groups. Psychology literature highlights religion as a potential coping mechanism (Koenig, 2012; Pargament, 1997). Life presents individuals with numerous negative shocks, both anticipated and unforeseen. Humans may use coping mechanisms to endure all sorts of adverse situations — geography and climate events such as extreme droughts and natural disasters, sickness and death, and civil conflict, among many others. These situations can affect us at very different stages in life. In this regard, religious groups can provide relief, offering psychological solace through shared beliefs, social bonds, and potential financial advantages — such as those derived from enhanced information exchange and informal insurance mechanisms inherent to the social fabric of religious networks (Ager and Ciccone, 2017; Bentzen, 2019, 2021; Dube et al., 2022).

With this in mind, this paper studies the effect of negative shocks experienced at different stages in life on individuals' religious expression. Using individual-level data from the Afro-barometer survey and leveraging the variation in age of exposure to drought episodes based on location, I estimate the impact of drought on reported religion and membership in religious groups in a sample of 37 African countries. My econometric specification exploits geographical and temporal variation in drought exposure with fixed effects accounting for country-year-of-birth and location-time differences, so that identification comes from comparing birth cohorts within the same locality and survey year. I find that individuals exposed to more severe droughts during their formative years are significantly more likely to belong to a religious group as adults:

a one-unit increase in average drought exposure during ages 12 to 17 raises the probability of religious group membership by 1.7 percentage points, an effect that is robust across bracket definitions, exposure measures, and inference methods, and that is concentrated in the teenage years, fading by the early twenties. The effect is present and precisely estimated only among respondents who demonstrably grew up where they are observed — residents of their ethnic homeland, or members of their locality’s ethnic majority — and formative-year drought does not predict who stays, ruling out selection into staying. In contrast, reported religious affiliation does not respond to exposure at any age. A birth-cohort panel shows the complementary short-run pattern: while sustained severe drought is under way, membership rises temporarily at every adult age — but only exposure in the formative years leaves a permanent mark.

This empirical observation — that exposure to droughts during the formative years predicts religious group membership later in life — aligns with extensive interdisciplinary research identifying adolescence as a uniquely formative period for the enduring shaping of beliefs, values, and social affiliations. Neuroscience research shows that the brain continues developing from puberty into the mid-20s, making experiences in these years unusually likely to have lasting effects (Blakemore, 2012; Casey et al., 2008; Giedd et al., 1999; Mills et al., 2014; Steinberg, 2005); psychologically, it is the stage of identity formation, in which individuals explore and commit to values, beliefs, and group attachments (Erikson, 1968; Marcia, 1966; Meeus et al., 2010); and sociologically, it is marked by a shift toward peer and community influence that makes collective responses to shared adversity powerful socializing forces (Nelson et al., 2005; Blakemore and Mills, 2014; Malti et al., 2021). A crisis such as a drought intensifies the need for meaning, coping, and material support, which religious groups provide through established frameworks, social bonds, and mutual aid (Koenig, 2012; Pargament, 1997); attachments formed under these conditions can persist as deeply ingrained preferences and coping mechanisms for future adversity (Thaler and Sunstein, 2008). This confluence of factors explains why shocks during the formative years can have a disproportionately strong and enduring impact on religious group membership.

This study contributes to the literature on economic development and religion, embedded in the study of the relationship between economic development and culture (Nunn, 2022, 2012; Algan and Cahuc, 2010; Alesina and Giuliano, 2010; Fernández and Fogli, 2009; Guiso et al., 2009, 2006). While the overall correlation between secularization and economic development has been thoroughly documented (McCleary and Barro, 2006), the direction of the theoretical causal relationship and the mechanisms through which this relationship has developed still remain

uncertain. The literature shows evidence supporting both directions of this link. Particularly, there is little evidence on factors determining people's religious adherence or practice decisions. Partly because it is often hard to quantify or measure appropriately.

On one hand, there is a growing literature linking religion and particular religious practices to economic development. Lawson (2018) documents generally that rises in secularization have preceded economic growth, while Andersen and Bentzen (2022) find that religiosity hindered the transition to modern growth. In terms of religious activities, there is evidence supporting that church attendance (McCleary and Barro, 2006), longer Ramadan fasting (Campante and Yanagizawa-Drott, 2015), and religious festivals that coincide with agricultural activity (Montero and Yang, 2022) are linked to lower development levels.

Parallely, when considering religion as a dependent variable, there is varied evidence arguing that a negative relationship exists between various measures of income, economic development, or economic activity and religion. Broadly speaking, McCleary and Barro (2006) finds that countries with higher income levels have lower religious adherence. While there are studies that find a positive relationship between income and religious adherence due to adversity testing people's faith (Buser, 2015) or time availability for religious practices (Auriol et al., 2020), most literature seems to point in the opposite direction, with arguments and evidence supporting that negative economic shocks of different nature increase religion or religious adherence (Dube et al., 2022; Bentzen, 2021; Henrich et al., 2019; Bentzen, 2019; Ager and Ciccone, 2017; Azzi and Ehrenberg, 1975). This paper adds to this literature by providing further evidence in line with the negative relationship between positive economic shocks and religion, by showing that individuals are more likely to belong to religious groups if they experience economic hardship in the form of exposure to droughts.

Furthermore, the mixed theoretical arguments and evidence suggest that it is necessary to analyze further the mechanisms through which this relationship operates. This paper expands on the literature focused on religion as a form of financial or emotional coping mechanism (Ager and Ciccone, 2017; Bentzen, 2019, 2021; Dube et al., 2022), and on the link between religion and the tightening of social norms in response to negative shocks (Henrich et al., 2019), by identifying the specific margin on which the response operates: droughts in the formative years raise membership in and contact with religious organizations, while leaving affiliation, stated religiosity, and the frequency of practice unchanged — a response of institutional belonging rather than of belief.

In addition to the above literature, this paper contributes to the economic literature on the heterogeneous effects of negative shocks across humans' life cycle. This literature is broad, exploring how shocks at different stages in life can impact a range of outcomes. The literature also includes papers exploring how early life conditions might shape later-life outcomes (Persico et al., 2004; Maccini and Yang, 2009; Yi et al., 2015), and how some particular shocks might have heterogeneous effects across the human life cycle (Salvanes et al., 2024). For instance, Persico et al. (2004) demonstrate that adolescent experiences, specifically teen height, have a lasting impact on labor market outcomes, with the wage premium for taller individuals being largely determined by their height during teenage years, mediated through participation in high school social activities like sports and clubs. This paper is the first to document the fact that the link between these shocks and religiosity occurs mostly at earlier stages in life.

Finally, this paper makes an empirical contribution to the identification of long-run effects of climate shocks in surveys without migration histories (Dell et al., 2014). Combining the Murdock ethnographic atlas with the geocoded survey locations, I construct two independent proxies for having grown up in place and show that the formative-years effect belongs to non-migrants — an approach applicable to any setting where lifetime exposure must be inferred from current location.

This paper continues as follows. First, it describes the context of the study and the data used in Sections 2 and 3. Discussion of the empirical approach can be found in Section 4. Section 5 reports the main results and the life-cycle profile of the effect. Section 6 shows that the results are not driven by migration. Section 7 examines what kind of response the effect represents and rules out several competing channels, and Section 8 collects robustness checks. Finally, Section 9 concludes.

2. Religion and Droughts in Africa

2.1. Religion

Religion shapes societal norms and cultural frameworks, significantly influencing human behavior and decision-making processes. The empirical findings emphasized by McCleary and Barro (2006) underscore a widely known pattern: religion tends to be more prevalent in developing nations than in their developed counterparts. Notably, within the Christian community, regions such as sub-Saharan Africa, Latin America, and the United States exhibit heightened religious

significance, while strong religious adherence among Muslims is observed in Africa, the Middle East, and South Asia.

Across the African continent, the distribution of religions reflects a diverse and dynamic landscape, with Islam and Christianity emerging as the two dominant faiths, alongside various indigenous and traditional belief systems (Center, 2010). Islam holds a significant presence across North Africa, the Sahel region, and parts of East Africa, with countries like Algeria, Egypt, Nigeria, and Sudan hosting sizable Muslim populations. Conversely, Christianity has established strong footholds in Sub-Saharan Africa, particularly in countries such as Nigeria, Ethiopia, South Africa, and the Democratic Republic of the Congo (Thornton, 2000). Many African nations exhibit a rich tapestry of indigenous religions, encompassing diverse spiritual practices, ancestral worship, and cultural traditions. While Islam and Christianity often constitute the primary religious affiliations, these traditional belief systems continue to influence local customs, rituals, and social structures, contributing to the religious pluralism and syncretism observed across the continent.

These diverse religious landscapes also feature distinct rites of passage that occur during childhood and adolescence, periods critical for identity and belief formation. For Muslims, male circumcision (*khitan*) is a widespread practice, often performed between the ages of 7 and puberty, marking a boy's transition towards adulthood (Wensinck, 1986). In Catholic Christianity, sacraments like Confirmation typically occur in the teenage years, signifying a personal affirmation of faith and a transition to youth or adulthood (Catholic Church, 1983). Protestant denominations often practice believer's baptism, a public declaration of faith, around the early teenage years or later, while confirmation, affirming personal commitment, also frequently takes place during adolescence (Isichei, 1995). Similarly, traditional African religions feature significant puberty or initiation rites during adolescence, typically between ages 12 and 18, which serve to mark the transition from childhood to adulthood and impart essential knowledge and skills for adult life and community roles (van Gennep, 1960; Mbiti, 1969; Turner, 1967). These rituals, occurring during formative years, play a crucial role in solidifying religious identity and group affiliation.

Thus, Africa emerges as a unique region where varied and diverse religions hold considerable importance — religions which also involve important and diverse transitional stages through childhood and adolescence. This single fact underscores the necessity to deepen our understanding of not only the causal relationship between religion and development but also the underlying mechanisms driving this association. Since a significant proportion of the world's

poorest nations are concentrated in Africa (The World Bank, 2022), understanding the dynamics of religion assumes critical significance in the process of elucidating the intricate interplay between religion and economic development.

2.2. Droughts

When considering the economic development of countries in Africa, both historically and today, geography — resource endowments, such as land and climate factors, among others — have long been recognized as relevant in explaining the difficulties faced by many countries in the region for achieving sustained economic development through varied mechanisms (Alesina and Giuliano, 2010; Alsan, 2015; Bloom et al., 1998; Sokoloff and Engerman, 2000). Among these factors, drought exerts profound and multifaceted impacts on Africa, rendering them a critical concern for the continent's socio-economic development and resilience to climate change.

Globally, drought is one of the most lethal hazards (World Meteorological Organization (WMO), 2021), mainly because it increases food insecurity, which brings with it a cascade of highly damaging impacts. Due in part to climate change, incidents of drought have doubled in the last 40 years, and their geographic range has expanded (for Disaster Risk Reduction, 2023). This has reversed gains in food security and poverty reduction (FAO, n.d.a.). Between 2014 and 2020, the prevalence of moderate to severe food insecurity rose by 22% to 30% with the highest rises in Sub-Saharan Africa, Central and South Asia, and Latin America (FAO, n.d.b.). Between 1970 and 2019 Africa accounted for 34% of all reported drought events globally. The Sahel region experienced extreme droughts in the 1970s and 1980s, significantly impacting food security and livelihood. For instance, the drought in the early 1980s affected over 50 million people in this region alone. Moreover, in Southern Africa, the 2015-2016 El Niño event led to widespread drought, affecting millions of people and leading to crop failures and water shortages (United Nations Office for Disaster Risk Reduction, 2019).

The region's heavy reliance on rain-fed agriculture makes it particularly vulnerable to the adverse effects of droughts, which can lead to crop failures, food shortages, and heightened economic instability. Moreover, drought-induced water scarcity exacerbates challenges in accessing clean water for drinking, sanitation, and hygiene, further exacerbating health risks and socio-economic disparities (Field et al., 2014). Thus, the region's rich tapestry of religions and vulnerability to droughts highlight the importance of understanding the dynamics between the two, and their possible link to economic development.

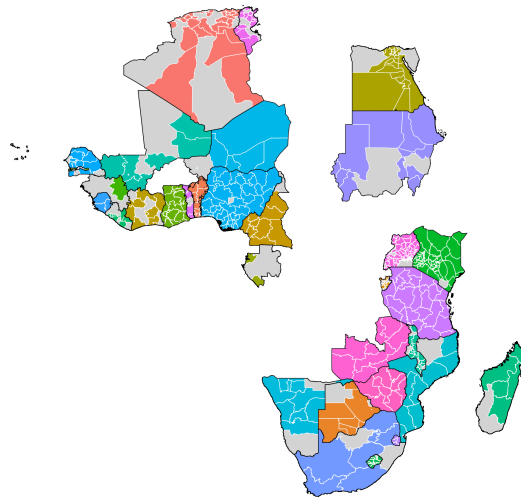
3. Data

To study the effect of drought exposure on religious decisions, I combine information on individuals' religion, measures of different expressions of religiousness or religious group membership, location, and demographic characteristics — particularly date of birth — from the Afrobarometer with spatial time series climate information that allows me to measure drought exposure in the region throughout the years.

3.1. Religious and Demographic Characteristics

This paper utilizes individual-level data from Rounds 5, 6, and 7 of the Afrobarometer survey, covering the period circa 2011 - 2018. From this dataset, I use information on individuals' age, employment status, date of survey, geographic location, sex, race, ethnicity, reported religion, and religious group membership status. The latter are the dependent variables of interest: *Religious* and *Member* — binary variables that take a value of one if the individual is religious or a member of a religious group, and zero otherwise. The final harmonized sample for the analysis includes 145,979 observations from 599 sub-national units, referred to as *localities*, spanning 37 African countries. For a detailed explanation of the Afrobarometer survey, specific variable definitions, country coverage, and harmonization procedures, please refer to Appendix A.1.

Figure 1: Countries and Localities in the Sample



Notes: Country borders denoted with black, and locality borders with white. The sample includes all localities that appear at least twice across all Afrobarometer rounds included, determined by the location in the Afrobarometer data (Data) in combination with GADM shapefiles from the Global Administrative Areas database (Global Administrative Areas, 2022). Missing localities are colored in light grey.

3.2. Drought Measurement

To measure drought exposure and intensity, I utilize The Global SPEI Database, which provides the Standardized Precipitation Evapotranspiration Index (SPEI-12). This index accounts for the joint effects of temperature and precipitation on droughts over a rolling 12-month period. Following Vicente-Serrano et al. (2010), I use SPEI-12 to characterize drought events. The SPEI scale defines categories such as Slight, Moderate, Severe, and Extreme drought, and is measured in standard deviations. To identify positive increases as increases in drought, I convert negative drought values — which represent lack of water — to positive.

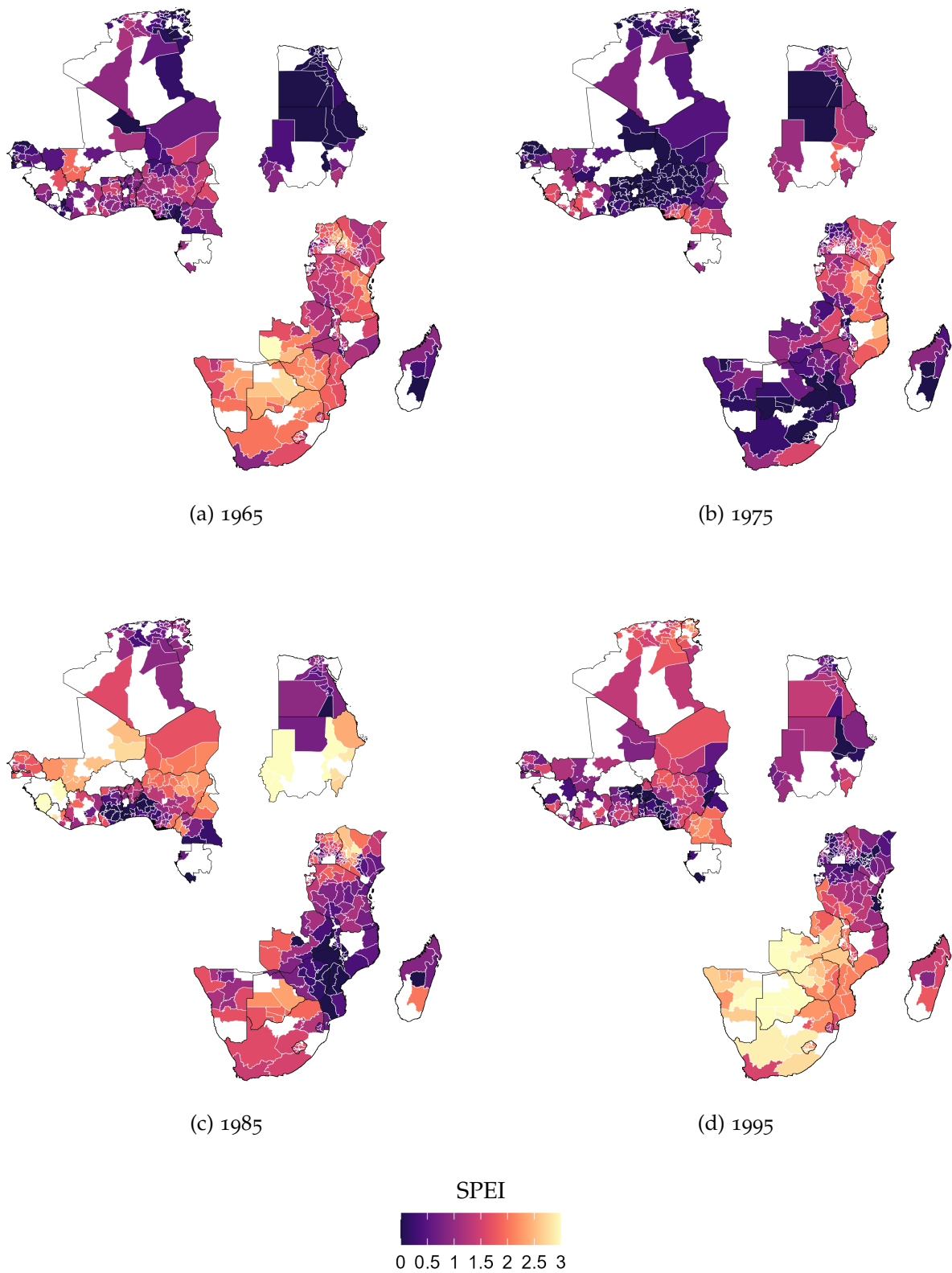
Figure 2 illustrates the data for drought levels for the relevant countries and locations across the years, highlighting that drought exposure varies greatly across locations and time, providing key identifying variation for the analysis.

For the main regression analysis, I construct $\text{Avg}(\text{Drought Exposure})$, which represents the average cumulative drought exposure from an individual's birth year until the survey year, based on their location. Additionally, to explore the life-cycle component, I construct a cumulative drought measure for six distinct age brackets. For a full description of the SPEI database, drought categories, and detailed variable construction, please refer to Appendix A.2.

3.3. Additional Data Sources

To account for possible migration, I incorporate Murdock (1959)'s ethnolinguistic map for the African continent — fortified with data from Murdock (1967)'s Ethnographic Atlas linked to the Human Relations Area Files. I match respondents' reported ethnicity to atlas homelands and use the homeland polygons in two ways: to measure drought exposure at the ethnic homeland using the same procedure as in Section 3.2, and to construct the first of two proxies for having grown up in place — an indicator for residing within 50 kilometers of one's ethnic-homeland polygon, which holds for 63% of matched respondents. The second proxy uses the survey's sampling structure: the share of respondents in an individual's geocoded locality who report the same ethnicity, with local majorities being overwhelmingly natives. Finally, for the moderator analysis in Section 8, I use country GDP per capita at the time of exposure from the Maddison Project Database (Bolt and van Zanden, 2024) and the long-run drought frequency of each locality computed from the SPEI record.

Figure 2: Overview of Drought Conditions from 1965 to 1995



Notes: Country borders denoted with black, and locality borders with white. The sample includes all localities that appear at least twice across all Afrobarometer rounds included, determined by the location in the Afrobarometer data (Data) in combination with GADM shapefiles from the Global Administrative Areas database (Global Administrative Areas, 2022). Missing localities are colored in white. SPEI index represented in reversed values, with positive numbers describing droughts.

3.4. Summary Statistics

In Table 1, I present summary statistics of the main variables used in the regression analysis. Relatively consistent with the age distribution across these countries, the average age — among those aged 18 and older — is around 37 years old, with a standard deviation of 14.4. As expected, the sample is evenly divided among men and women. Only 36% report being employed at the time of the survey. Around 81% of the sample report some level of formal education, but only 15% report having post-secondary education.

Table 1: Summary Statistics of the Sample

	Mean	SD	Median	Min	Max	N
Age	37.01	14.41	34.00	18.00	100.00	145,979
Male	0.50	0.50	0.00	0.00	1.00	145,979
Formal Education	0.81	0.39	1.00	0.00	1.00	145,546
Primary Education	0.30	0.46	0.00	0.00	1.00	145,546
Secondary Education	0.36	0.48	0.00	0.00	1.00	145,546
Post-secondary Education	0.15	0.35	0.00	0.00	1.00	145,546
Employed	0.36	0.48	0.00	0.00	1.00	145,440
Religious	0.97	0.18	1.00	0.00	1.00	145,979
Muslim	0.32	0.47	0.00	0.00	1.00	145,979
Christian	0.59	0.49	1.00	0.00	1.00	145,979
Other	0.06	0.23	0.00	0.00	1.00	145,979
None	0.03	0.18	0.00	0.00	1.00	145,979
Religious Group Member	0.47	0.50	0.00	0.00	1.00	145,979
<i>Avg(Exp)</i>	1.27	0.30	1.26	0.32	2.46	145,979

In terms of the variables related to religion, 32% report being Muslim, while 59% report being Christian, 6% say they have another religion, and only 3% report having no religion. The individuals in the survey report high membership levels in religious groups, with 47% reporting being a member and 31% an active member.

Regarding the drought measurements, we can see that — on average — individuals have an exposure of 1.27 of cumulative drought per year, equivalent to a moderate drought, based on the SPEI scale.

4. Identification Strategy

The identification strategy comes from the quasi-exogenous variation in exposure to droughts for individuals based on their age and location. That is, depending on the location and year when the individual was born, individuals at any point in time might have experienced more or fewer droughts, higher or lower intensity of droughts, and — more importantly — experienced these drought episodes at different stages in life. Recognizing that individuals born in different years in different countries have other inherent differences that might confound the analysis, I propose the following regression analysis:

$$y_{i(c,k,j,t)} = \alpha + \beta f\left(\mathbf{D}_{i(c,k,j,t)}\right) + \gamma \mathbf{X}_i + \delta_{j(i),t(i)} + \phi_{c(i),k(i),t(i)} + \varepsilon_{i(c,k,j,t)} \quad (1)$$

where $y_{i(c,k,j,t)}$ is the religiosity measure for individual i (from age-cohort c , location j , country k , year t). $\mathbf{D}_{i(c,k,j,t)}$ represents the drought exposure measure for individual i in location j in year t . $\mathbf{X}_i$ includes individual-level controls; the main specification controls for gender only, because characteristics such as education, employment status, urban residence, and religion are measured in adulthood and are themselves potential outcomes of formative-year exposure, so conditioning on them would introduce post-treatment bias. $\delta_{j(i),t(i)}$ denotes Locality-Year Fixed Effects to capture time-varying location-specific differences, and $\phi_{c(i),k(i),t(i)}$ represents Year of Birth Cohort-Country-Year Fixed Effects, included to account for the inherent differences across individuals born in different years in different countries observed in different rounds. That is, the variation in $y_{i(c,k,j,t)}$ and $\mathbf{D}_{i(c,k,j,t)}$ comes from variation in an individual's birth-year cohort c , sub-national location j , and year of observation t . Finally, $\varepsilon_{i(c,k,j,t)}$ represents the idiosyncratic error term, clustered at the country-year level; Section 8 shows the results are unchanged under clustering by locality, two-way clustering, and Conley (1999) spatial HAC standard errors.

The religiosity measures $y_{i(c,k,j,t)}$ of interest are two binary variables: *Religious*, which takes a value of 1 if the individual reports having a religion — Muslim, Christian, or other — or 0 otherwise; and *Member*, which takes a value of 1 if the individual is a member of a religious group and 0 otherwise.

The coefficient of interest in Equation (1) is β , the effect of a one-unit increase in average yearly drought exposure — one standard deviation on the SPEI scale, a variation that pushes individuals from no drought to slight drought, slight to moderate, and moderate to severe — on the outcome of interest (e.g. probability of being member of a religious group). When examining

the impacts on these religiosity measures, the hypothesis is that $\beta > 0$, that is, exposure to more severe droughts increases the probability of individuals participating in religious groups.

Including the function $f(x)$ in Equation (1) allows for capturing the relationship between the drought measure $\mathbf{D}_{i(c,k,j,t)}$ and the outcome $y_{i(c,k,j,t)}$ in a flexible manner. The main specification in this paper focuses on either (i) the yearly average drought exposure that individuals have experienced through their lifetime, $Avg(Exp)$, or (ii) a set of variables calculating the same yearly average but for different age brackets $\sum_{l=1}^L Avg(Exp_l)$. These functional forms enable the estimation of both the overall aggregate effect of drought exposure on religiosity and religious expression measures, as well as distinguishing the differential impact that exposure might have throughout different stages in life.

The main identifying assumption is that $E[\varepsilon_{i(c,k,j,t)} | \mathbf{D}_{i(c,k,j,t)}] = E[\varepsilon_{i(c,k,j,t)}] = 0$, that is, the drought exposure measure for individual i — from age-cohort c , location j , country k , year t — is uncorrelated with omitted variables that may also affect religiosity and religious expression conditional on both Locality - Year and Birth Cohort - Country - Year fixed effects.

4.1. Threats to Identification

The identifying variation is the interaction of birth cohort and location: within a locality and survey year, cohorts who were adolescents during local drought years are compared to cohorts who were not. Because the data record only *current* location, the central threat is migration — exposure is measured as if each individual had always lived in their surveyed locality — and Section 6 is devoted to showing that this threat does not drive the results.

Two further threats concern the cohort dimension of the variation. First, the cohort-country-year fixed effects absorb anything common to a birth cohort within a country, but not slow-moving, locality-specific differences across successive cohorts: if the localities that happened to be dry while a given cohort was adolescent were also localities where, say, missionary activity or schooling was expanding cohort by cohort, the exposure brackets could pick up such trends. Section 8.3 addresses this in two ways. It re-estimates the model with locality-specific linear trends in year of birth, and it uses schooling — which is completed by the early twenties and therefore cannot respond to exposure after that age — as a placebo outcome to test whether the design generates spurious coefficients on late-life brackets. Second, the sample consists of survivors: if mortality or emigration out of the sampling frame responded to drought, the

cohorts observed in a locality would be selected on their exposure. Appendix B.7 shows that formative-year exposure does not predict the number of respondents in a locality-cohort cell, that the estimate is stable when any single country or any decade of adolescence is dropped, and that it holds in balanced cohort ranges; it also reports randomization inference that reassigns entire drought histories across localities within a country.

5. Empirical Results: Religiousness & Religious Group Membership

This section presents the empirical estimates of the impact of droughts on religiousness and religious group membership, discussing the heterogeneous effects that drought exposure has throughout the life cycle.

5.1. Droughts and Religion

Examining the impacts of drought exposure on religiousness and religious expression, Table 2 presents the estimates for Equation (1), with religiousness (columns (1) and (2)) and religious group membership (columns (3) and (4)) as the dependent variables. Columns (1) and (3) represent the estimates of Equation (1) using $Avg(Exp)$ — average drought exposure through an individual’s lifetime — as the drought exposure measurement $\mathbf{D}_{i(c,k,j,t)}$, while columns (2) and (4) estimate Equation (1), where $\mathbf{D}_{i(c,k,j,t)}$ represents the average drought exposure for the six different age brackets $\sum_{l=1}^l Avg(Exp_l)$.

Table 2: Impact of Droughts on Religiousness and Group Membership

	Dependent Variable:			
	Religious		Member	
	(1)	(2)	(3)	(4)
<i>Avg(Exp)</i>	-0.005 (0.011)		0.052* (0.028)	
<i>Avg(Exp₀₋₅)</i>		-0.001 (0.003)		-0.003 (0.008)
<i>Avg(Exp₆₋₁₁)</i>		-0.000 (0.003)		0.007 (0.007)
<i>Avg(Exp₁₂₋₁₇)</i>		-0.003 (0.003)		0.017** (0.006)
<i>Avg(Exp₁₈₋₂₃)</i>		0.002 (0.003)		0.011* (0.006)
<i>Avg(Exp₂₄₋₂₉)</i>		0.000 (0.002)		0.009 (0.007)
<i>Avg(Exp₃₀₊)</i>		-0.003 (0.003)		-0.007 (0.007)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Observations	145,979	145,979	145,979	145,979
Num Localities	599	599	599	599
Num Countries	37	37	37	37
Adj R^2	0.084	0.084	0.187	0.187
Outcome Mean	0.965	0.965	0.465	0.465
Outcome SD	0.184	0.184	0.499	0.499

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. *Religious* takes a value of 1 if the individual reports being religious and 0 otherwise. *Member* takes a value of 1 if the individual reports being a member of a religious group and 0 otherwise. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Results reveal that, while drought exposure has no effect on individuals' reported religiosity at any point of the life cycle, it has a positive and significant effect on the probability of being a member of a religious group. A one-unit increase in lifetime average drought exposure — a substantial move along the SPEI drought scale, comparable to shifting from one drought category to the next — is associated with a 5.2 percentage point increase in the probability of reporting religious group membership, approximately 11% of the dependent variable mean. Equivalently, a one standard deviation increase in exposure (0.30 units) raises the probability of membership by roughly 1.6 percentage points.

Focusing on the age-bracket estimates, Table 2 shows that this aggregate effect is concentrated in the formative years. The coefficient on exposure during ages 12 to 17 is 0.017 ($p < 0.05$) — 0.025 ($p < 0.01$) once locality-specific cohort trends are added (Section 8.3) — with a smaller and marginally significant coefficient at ages 18 to 23 (0.011, $p < 0.10$), while exposure in early

childhood (ages 0–11) and in later adulthood (ages 30 and above) has no effect. The brackets are jointly significant for membership ($p = 0.021$), and the non-teenage brackets are jointly indistinguishable from zero ($p = 0.108$). The horse race in Table 3 sharpens this comparison: pitting the teenage window directly against the pooled exposure at all other ages, the teenage coefficient is precisely estimated at 0.020 ($p < 0.01$), while the pooled other-age exposure is of similar magnitude but statistically indistinguishable from zero. The response is not confined to nominal belonging: the effect is at least as strong on the intensive margin — active membership, including leadership roles — which is examined alongside membership from Section 6 onward.

Table 3: Horse Race: Teenage-Window vs. Other-Age Exposure

	Dependent Variable:	
	Religious	Member
	(1)	(2)
$Avg(Exp_{12-17})$	−0.002 (0.003)	0.020*** (0.006)
$Avg(Exp_{other})$	−0.000 (0.009)	0.022 (0.020)
L x Y FE	Y	Y
C x Y x YOB FE	Y	Y
$p(\text{all brackets}=0)$	0.679	0.021
$p(\text{non-teen brackets}=0)$	0.747	0.108
$p(\text{teen}=other)$	0.832	0.923
Observations	145,979	145,979
Num Localities	599	599
Num Countries	37	37
Adj R^2	0.084	0.187
Outcome Mean	0.965	0.465
Outcome SD	0.184	0.499

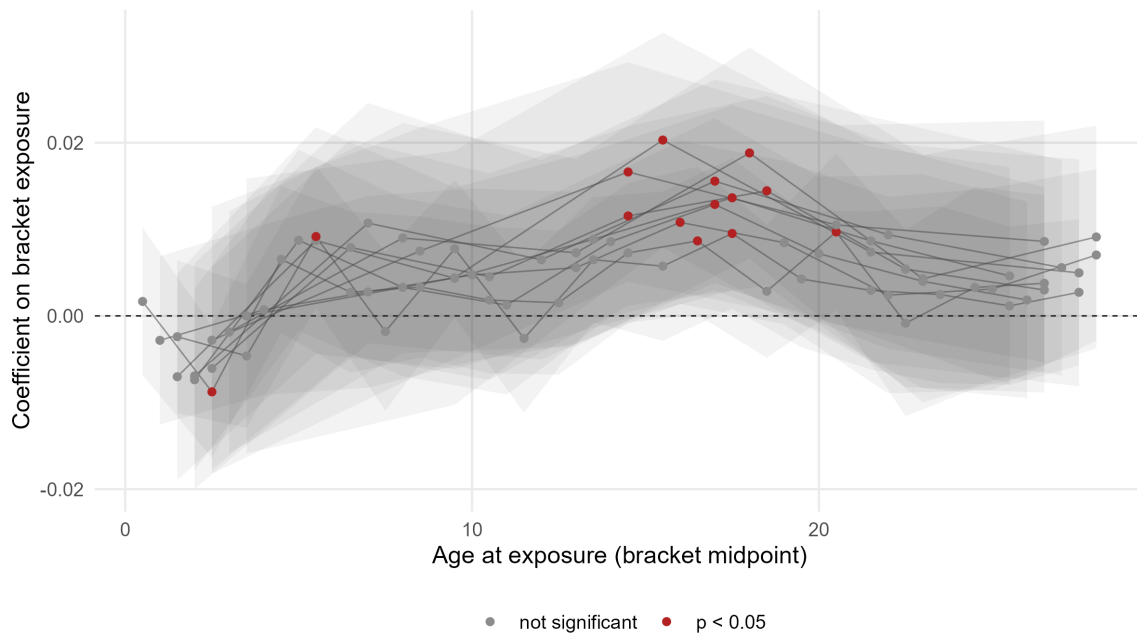
Notes: Each column reports a two-regressor specification in which $Avg(Exp_{12-17})$ is the average drought exposure during the teenage window (ages 12–17) and $Avg(Exp_{other})$ is the equal-weighted mean of exposure across all other age brackets. The rows “ $p(\text{all brackets}=0)$ ” and “ $p(\text{non-teen brackets}=0)$ ” report joint F -tests from the full age-bracket model in Table 2. Standard errors clustered by Country-Year in parentheses. Controlling for gender. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2. The Life-Cycle Profile of the Effect

The 0/6 age brackets used above were fixed *a priori*, and the concentration of the effect in the 12–17 bracket could in principle be sensitive to that choice. Figure 3 therefore re-estimates Equation (1) under every bracketing scheme with widths of two to six years and starting ages of zero or one — ten schemes and 83 bracket coefficients in total. The teenage window is not an artifact of the grid: coefficients on brackets covering ages younger than 12 are uniformly close to zero, the effect

emerges for brackets entering adolescence, peaks for brackets centered around ages 15 to 19, and fades for brackets beyond the late twenties.

Figure 3: Age-Bracket Sweep: Drought Exposure and Religious Group Membership



Notes: Each line traces the bracket coefficients from one estimation of Equation (1) with *Member* as the dependent variable, under one of ten age-bracketing schemes (widths 2–6 years, starting ages 0 or 1; residual brackets excluded). Points mark bracket coefficients at the bracket midpoint; red points are significant at the 5% level. Shaded areas are 95% confidence intervals.

The profile the sweep traces is an inverted U over the life cycle: nothing for childhood brackets, a peak for brackets centered on ages 15–19, and decay for brackets reaching beyond the mid-twenties. Section 6 returns to this profile with a smoothed per-year estimate and developmental windows, once the sample in which exposure is best measured has been identified.

Heterogeneity by religion and gender is examined in Appendix B.11: the formative-years pattern on membership holds across all three religious groups, and is concentrated among men.

6. Growing Up in Place: Ruling Out Migration

The Afrobarometer does not record migration histories, so the exposure measure assigns each respondent the drought history of their current locality. The estimates above could therefore reflect sorting — adults who experienced drought in their youth moving differentially into the sampled localities — rather than the lasting effect of a lived childhood environment. This section

shows that the opposite is true: the effect is present, and precisely estimated, only among people who demonstrably grew up where they live, and never appears among likely migrants.

I construct two independent proxies for having grown up in place. The first uses the ethnographic record: a respondent whose locality lies within 50 kilometers of their matched Murdock ethnic-homeland polygon cannot have migrated over a long distance away from their group's ancestral area; 63% of matched respondents satisfy this criterion.¹ The second requires no ethnographic match: respondents whose ethnic group constitutes the majority of their own survey locality are overwhelmingly natives, while local minorities are disproportionately in-migrants. The two proxies agree — local majorities reside in their homeland 78% of the time, local minorities 39% — but they fail in different ways, which makes their agreement in the regressions informative.

Table 4 estimates the age-bracket specification separately for stayers and movers under each proxy. The teenage effect on membership among homeland residents is 0.030 ($p < 0.01$) — nearly twice the full-sample estimate — and 0.025 ($p < 0.01$) among local ethnic majorities, while it is an order of magnitude smaller and statistically indistinguishable from zero among respondents living outside their homeland or in localities where they are an ethnic minority. This is exactly the pattern implied by measurement: for movers, current-location exposure mismeasures the childhood environment and the estimate attenuates toward zero. It is the opposite of what sorting would produce, which requires the effect to be carried by movers. Because the mover estimates are imprecise, the formal test of equality of the teen coefficients across groups — reported in the table from a fully interacted model — is not sharp; the force of the evidence lies not in that contrast but in the effect being present and precisely estimated exactly where exposure is well measured, under two proxies that fail in different ways, and never among movers.

¹This share is close to what direct migration histories imply for the same region: applying the same ethnicity-to-homeland matching to geocoded Demographic and Health Survey data for 20 African countries, in which residential histories are recorded, 66% of respondents with an identified homeland live within 50 kilometers of it (author's calculation in companion work; not reported here).

Table 4: Drought Exposure and Membership: Stayers versus Movers

	Ethnic Homeland		Locality Ethnic Share	
	In homeland	Outside	Majority	Minority
$Avg(Exp_{0-5})$	0.005 (0.011)	-0.018 (0.016)	0.007 (0.010)	-0.013 (0.014)
$Avg(Exp_{6-11})$	0.008 (0.011)	0.016 (0.015)	0.011 (0.010)	-0.001 (0.014)
$Avg(Exp_{12-17})$	0.030*** (0.010)	0.003 (0.013)	0.025*** (0.009)	0.002 (0.012)
$Avg(Exp_{18-23})$	0.017* (0.009)	0.013 (0.014)	0.006 (0.007)	0.027** (0.012)
$Avg(Exp_{24-29})$	0.017* (0.010)	0.005 (0.015)	0.016* (0.008)	0.001 (0.014)
$Avg(Exp_{30+})$	-0.003 (0.011)	-0.008 (0.014)	0.000 (0.009)	-0.017 (0.013)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
p(all brackets=0)	0.018	0.686	0.070	0.157
p(teen effect equal across split)	0.128	0.128	0.136	0.136
Observations	70,224	40,620	85,916	51,526
Num Localities	462	458	570	512
Num Countries	30	31	36	34
Adj R^2	0.158	0.148	0.183	0.150
Outcome Mean	0.519	0.475	0.480	0.485
Outcome SD	0.500	0.499	0.500	0.500

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Dependent variable: *Member*. Columns (1)–(2) split the Murdock-matched sample by whether the respondent lives within 50 km of their ethnic-homeland polygon; columns (3)–(4) split the full sample by whether the respondent belongs to the ethnic majority of their geocoded survey locality (share ≥ 0.5). “p(teen effect equal across split)” tests equality of the two displayed teenage coefficients in a fully interacted model — group-specific bracket slopes, controls, and fixed effects — so it corresponds to the columns shown. The corresponding estimates for *Religious* are null in all four columns (Appendix B.4). Results are robust to homeland-distance cutoffs of 25 and 100 km, to ethnic-share cutoffs of 0.3 and 0.7, and to restricting to countries with both stayers and movers (Appendix B.3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A remaining concern is that being a stayer is itself an outcome: if formative-year drought changed who remains in place, the stayer subsample would be selected on the treatment. Table 5 tests this directly by regressing each residence proxy on the full set of exposure brackets. Teenage drought exposure predicts none of them — neither homeland residence, nor distance to the homeland, nor local ethnic composition — with coefficients of at most 0.007 and joint tests far from rejection in three of four columns. The only significant coefficients anywhere in the table are for exposure at ages 6–11 — on homeland residence and on log distance to the homeland, two mutually consistent readings of the same movement — an age window in which membership shows no response, so they cannot account for the results. The stayer split, in other words, does not condition on the treatment.

A subtler version of the concern is that the 50-kilometer criterion still allows local moves — in particular, from a village to a town within the homeland area. Such moves barely affect

measurement, since drought is regional and a nearby town shares the locality's drought history, but they could constitute a channel: if drought pushed adolescents into towns, where formal religious organizations are denser, the effect would partly reflect drought-induced urbanization rather than a formed attachment. Appendix Table B6 tests this directly: formative-year drought does not predict urban residence, in the full sample or among stayers (teenage coefficients of 0.003 and 0.011, jointly insignificant). Consistent with this, the effect is present among respondents currently working in agriculture — who by construction did not relocate to town — at 0.040 ($p < 0.05$) in the teenage window (Table B15).

Table 5: Selection Test: Does Formative-Year Drought Predict Staying?

	In homeland	ln(1+km to homeland)	Own-ethnic share	Local majority
<i>Avg(Exp₀₋₅)</i>	-0.002 (0.007)	0.031 (0.034)	-0.001 (0.004)	0.002 (0.006)
<i>Avg(Exp₆₋₁₁)</i>	0.012** (0.006)	-0.082*** (0.029)	-0.000 (0.004)	0.004 (0.006)
<i>Avg(Exp₁₂₋₁₇)</i>	0.002 (0.006)	0.006 (0.032)	0.001 (0.004)	0.007 (0.006)
<i>Avg(Exp₁₈₋₂₃)</i>	-0.001 (0.006)	-0.011 (0.031)	-0.002 (0.004)	-0.001 (0.006)
<i>Avg(Exp₂₄₋₂₉)</i>	-0.004 (0.006)	0.031 (0.028)	0.004 (0.004)	0.006 (0.006)
<i>Avg(Exp₃₀₊)</i>	-0.006 (0.007)	0.030 (0.037)	0.006 (0.005)	0.009 (0.006)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
p(all brackets=0)	0.362	0.037	0.710	0.729
Observations	111,837	111,837	138,507	138,507
Num Localities	503	503	575	575
Num Countries	31	31	36	36
Adj R^2	0.341	0.377	0.488	0.369
Outcome Mean	0.630	2.425	0.604	0.623
Outcome SD	0.483	2.443	0.341	0.485

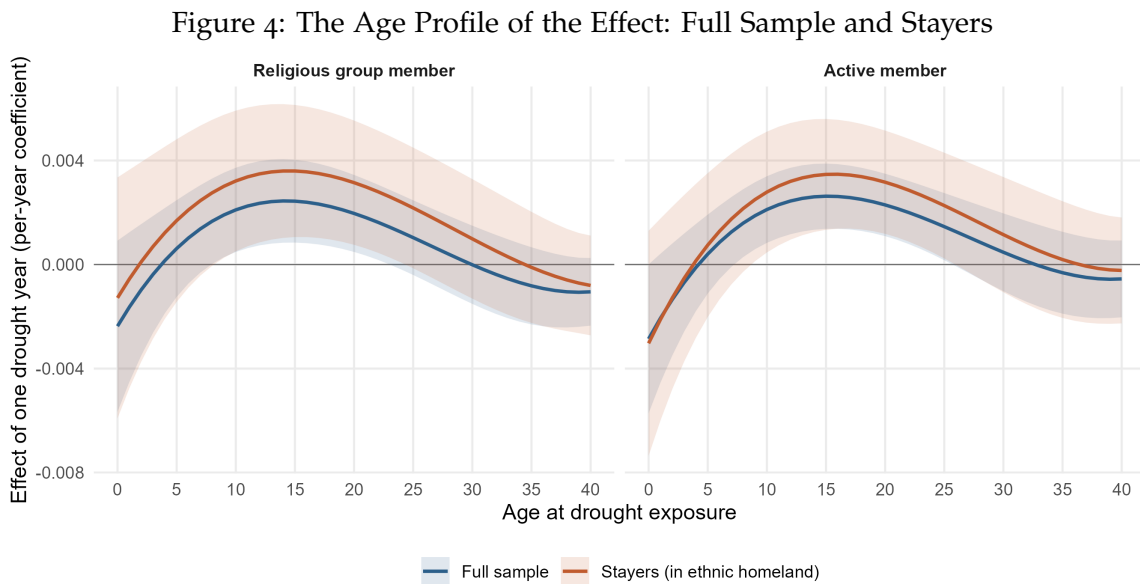
Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Each column regresses a residence proxy on the full set of drought-exposure brackets in the main estimation sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, Appendix B.1 reports the exposure measure computed at the ethnic homeland itself, which is mechanically immune to lifetime migration. Its teenage coefficient is null — both in the full matched sample and with ethnic-homeland fixed effects absorbing cross-group composition (Appendix B.2) — which is what the stayers results imply it should be: the homeland polygon is a coarse spatial proxy for the childhood environment (a typical polygon spans several localities), and a third of matched respondents did not grow up there. Consistent with this, among homeland stayers — for whom the homeland polygon and the current locality overlap — the homeland-based teenage coefficient is positive at 0.013 (s.e. 0.009; 0.014 with homeland fixed effects), about half the 0.030 obtained from current-location exposure in the same sample, and it

is negative among movers (Appendix Table B14).² The evidence thus consistently points to the lived childhood environment, not to ancestral climate or to sorting.

6.1. The Life-Cycle Profile Where Exposure Is Best Measured

With the stayers identified, the timing of the effect can be traced on the sample where exposure is best measured. Figure 4 estimates the effect of a single drought year at each age of exposure, smoothing the per-year coefficients with a third-degree polynomial in age, for the full sample and for homeland stayers. The two curves draw the same shape: no effect of childhood exposure, a rise through early adolescence to a peak at ages 14–16, decay through the twenties, and nothing after roughly age 30. The stayer curve is about 1.5 times the full-sample curve throughout — measurement error attenuates the profile, it does not reshape it. The bracket-scheme sweep of Figure 3, repeated on the stayer sample, shows the same concentration: no significant bracket below a midpoint of age 12, and significance clustered at midpoints 12–19 (Appendix B.9).



Notes: Each curve plots the effect of one additional drought year experienced at each age, from a single estimation of Equation (1) in which the per-year exposure coefficients are restricted to a third-degree polynomial in age (Almon distributed lag). Shaded areas are 95% confidence intervals; standard errors clustered by Country-Year. “Stayers” restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon.

²Consistent with this reading, applying the same matching to geocoded Demographic and Health Survey data with recorded residential histories, homeland exposure and current-location exposure yield the same estimates for non-migrants living inside their homeland, where the two measures largely coincide; for non-migrants living outside it, where the measures diverge and the comparison would be informative, neither is estimated precisely enough to adjudicate (author’s calculation in companion work; not reported here).

Table 6 summarizes the same profile in developmental windows — childhood (0–11), early adolescence (12–15), the transition to adulthood (16–20), early adulthood (21–29), and later life (30+). Because each coefficient multiplies *average* exposure over its window, it cumulates the per-year effects of the window’s years, and wider windows are therefore mechanically larger; Figure 4 is the unit-comparable version. Read this way, the windows tell one story: membership responds to exposure between roughly ages 12 and 20, the nine-year 21–29 window collects the decaying tail of the profile — its stayer coefficient reaches significance only because it pools nine years whose per-year effects are about half the adolescent peak, and four-year windows within it are individually insignificant — and childhood and post-30 exposure do nothing.

Table 6: Developmental Windows: Drought Exposure Across the Formative Years

	Member		Active Member	
	Full	Stayers	Full	Stayers
	(1)	(2)	(3)	(4)
$Avg(Exp_{0-11})$	0.005 (0.011)	0.013 (0.018)	−0.003 (0.010)	−0.000 (0.016)
$Avg(Exp_{12-15})$	0.009* (0.005)	0.022*** (0.008)	0.016*** (0.005)	0.020** (0.008)
$Avg(Exp_{16-20})$	0.018*** (0.006)	0.021** (0.009)	0.014** (0.006)	0.014 (0.010)
$Avg(Exp_{21-29})$	0.010 (0.008)	0.024** (0.011)	0.010 (0.007)	0.023** (0.010)
$Avg(Exp_{30+})$	−0.008 (0.008)	−0.003 (0.014)	−0.005 (0.009)	−0.001 (0.013)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
p(all windows=0)	0.004	0.004	0.000	0.009
p(12-15 = 16-20)	0.237	0.917	0.889	0.639
Observations	145,979	70,224	145,979	70,224
Num Localities	599	462	599	462
Num Countries	37	30	37	30
Adj R^2	0.187	0.158	0.175	0.165
Outcome Mean	0.465	0.519	0.307	0.346
Outcome SD	0.499	0.500	0.461	0.476

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. “Stayers” restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. “p(12-15 = 16-20)” tests equality of the early-adolescence and transition-to-adulthood coefficients. Window coefficients multiply average exposure within the window and therefore cumulate per-year effects; windows differ in width (see text). *Member* takes a value of 1 if the individual reports being a member of a religious group; *Active Member* additionally requires active membership or leadership. The corresponding estimates for (secular) community-group membership are reported in Appendix B.10. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7. Interpreting the Effect: What Kind of Response?

The results so far establish that formative-year drought durably raises membership in religious groups without changing religious affiliation. This section asks what kind of response that is — whether it reflects a general strategy of social attachment, a change in belief, or participation in religious institutions specifically — and rules out several competing channels.

Table 7 estimates the age-bracket specification for religious membership alongside the closest secular counterparts available in the survey: membership in voluntary community associations, and trust in local government councils and in traditional leaders. The contrast is stark. The same shock that raises religious membership by 1.7 to 3.0 percentage points at ages 12–17 moves community-group membership by less than 0.7 percentage points ($p > 0.4$) and moves trust in local and traditional authority not at all, with point estimates that are small and negative. The response is not a generalized turn toward association or authority; it is specific to religious institutions. The one qualification is that community groups show some response at ages 16–20 in the full sample — later and roughly half as strong as the religious response (Appendix B.10). But unlike religious membership, this response does not survive the stayer restriction: among homeland stayers the community-group windows are individually and jointly insignificant, whereas the religious-membership effect grows. The adolescent imprint that survives the migration filter is distinctly religious.

Table 7: Religion-Specific or General? Membership and Trust Outcomes

	Member		Active Member		Community Member		Trust Local Council		Trust Trad. Leaders	
	Full	Stay	Full	Stay	Full	Stay	Full	Stay	Full	Stay
<i>Avg(Exp₀₋₅)</i>	-0.003 (0.008)	0.005 (0.011)	-0.006 (0.006)	-0.007 (0.010)	0.001 (0.008)	-0.005 (0.012)	-0.002 (0.006)	0.006 (0.010)	0.002 (0.008)	0.008 (0.011)
<i>Avg(Exp₆₋₁₁)</i>	0.007 (0.007)	0.008 (0.011)	0.004 (0.007)	0.010 (0.010)	0.007 (0.007)	-0.010 (0.011)	-0.010 (0.007)	-0.002 (0.011)	-0.011 (0.008)	-0.013 (0.012)
<i>Avg(Exp₁₂₋₁₇)</i>	0.017** (0.006)	0.030*** (0.010)	0.022*** (0.006)	0.025*** (0.009)	0.007 (0.008)	0.003 (0.012)	-0.003 (0.008)	-0.009 (0.012)	-0.006 (0.009)	-0.007 (0.014)
<i>Avg(Exp₁₈₋₂₃)</i>	0.011* (0.006)	0.017* (0.009)	0.013** (0.006)	0.022** (0.009)	0.011 (0.007)	0.007 (0.014)	0.007 (0.007)	0.026** (0.012)	0.013 (0.009)	0.011 (0.011)
<i>Avg(Exp₂₄₋₂₉)</i>	0.009 (0.007)	0.017* (0.010)	0.007 (0.007)	0.015* (0.009)	0.005 (0.008)	0.013 (0.012)	0.002 (0.006)	0.009 (0.009)	0.000 (0.009)	0.000 (0.014)
<i>Avg(Exp₃₀₊)</i>	-0.007 (0.007)	-0.003 (0.011)	-0.002 (0.007)	0.003 (0.011)	-0.001 (0.007)	-0.001 (0.009)	-0.008 (0.008)	-0.002 (0.013)	0.008 (0.010)	0.011 (0.013)
p(all brackets=0)	0.021	0.018	0.000	0.002	0.472	0.785	0.523	0.251	0.255	0.375
Observations	145,979	70,224	145,979	70,224	144,723	69,715	135,651	65,499	80,638	42,748
Num Localities	599	462	599	462	599	462	598	461	548	451
Num Countries	37	30	37	30	37	30	37	30	33	29
Adj R ²	0.187	0.158	0.175	0.166	0.140	0.132	0.125	0.124	0.130	0.134
Outcome Mean	0.465	0.519	0.307	0.346	0.362	0.411	0.482	0.515	0.637	0.659
Outcome SD	0.499	0.500	0.461	0.476	0.481	0.492	0.500	0.500	0.481	0.474

Notes: Standard errors clustered by Country-Year in parentheses. L represents Location, Y represents Year, C represents Country, and YOB represents Year of Birth. Controlling for gender. "Stayers" restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. Trust outcomes are indicators for trusting the institution "somewhat" or "a lot". Trust in traditional leaders is available in Rounds 6–7 only. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8 then characterizes the religious response itself, using items available in subsets of the survey rounds: trust in and contact with religious leaders, frequency of religious practice, and the stated importance of religion. The pattern mirrors the affiliation results: what moves is

engagement with the institution — contact with religious leaders rises by 2.0 percentage points ($p < 0.05$) — while trust in religious leaders, weekly practice, and the stated importance of religion show no significant response. Formative-year drought appears to build ties to religious organizations and their personnel rather than intensifying belief or salience.

Table 8: The Religious Response: Leaders, Practice, and Salience

	Trust Relig. Leaders		Contact Relig. Leader		Contact Trad. Leader		Weekly Practice		Religion V. Important	
	Full	Stay	Full	Stay	Full	Stay	Full	Stay	Full	Stay
$Avg(Exp_{0-5})$	0.004 (0.007)	0.015 (0.009)	-0.008 (0.009)	-0.006 (0.013)	0.002 (0.010)	0.006 (0.013)	0.000 (0.010)	0.011 (0.014)	-0.001 (0.009)	0.007 (0.014)
$Avg(Exp_{6-11})$	-0.007 (0.006)	-0.014 (0.009)	0.008 (0.009)	-0.002 (0.012)	0.011 (0.008)	-0.006 (0.010)	-0.004 (0.013)	0.000 (0.023)	-0.004 (0.007)	-0.015 (0.010)
$Avg(Exp_{12-17})$	0.002 (0.007)	0.005 (0.010)	0.020** (0.010)	0.008 (0.014)	0.014* (0.008)	0.010 (0.011)	0.015 (0.011)	0.022 (0.014)	0.001 (0.006)	0.009 (0.008)
$Avg(Exp_{18-23})$	0.007 (0.009)	0.003 (0.010)	0.009 (0.007)	0.009 (0.014)	0.005 (0.008)	-0.007 (0.013)	-0.013 (0.013)	-0.006 (0.016)	0.003 (0.007)	0.005 (0.011)
$Avg(Exp_{24-29})$	-0.002 (0.008)	0.007 (0.012)	0.009 (0.009)	0.005 (0.014)	0.005 (0.008)	0.001 (0.011)	0.014 (0.010)	0.020 (0.015)	-0.002 (0.008)	0.009 (0.010)
$Avg(Exp_{30+})$	0.024*** (0.008)	0.021* (0.012)	0.013 (0.009)	0.012 (0.014)	0.003 (0.009)	-0.005 (0.011)	0.008 (0.012)	0.022 (0.017)	-0.005 (0.010)	-0.013 (0.014)
p(all brackets=0)	0.026	0.180	0.138	0.934	0.413	0.940	0.559	0.596	0.954	0.423
Observations	94,230	46,013	96,381	46,791	87,527	45,838	51,133	24,923	47,843	22,660
Num Localities	575	452	575	452	548	452	560	436	528	371
Num Countries	36	29	36	29	33	29	35	28	33	26
Adj R^2	0.110	0.105	0.125	0.118	0.172	0.163	0.140	0.149	0.068	0.074
Outcome Mean	0.730	0.765	0.409	0.452	0.316	0.357	0.752	0.765	0.875	0.878
Outcome SD	0.444	0.424	0.492	0.498	0.465	0.479	0.432	0.424	0.330	0.327

Notes: Standard errors clustered by Country-Year in parentheses. L represents Location, Y represents Year, C represents Country, and YOB represents Year of Birth. Controlling for gender. "Stayers" restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. Trust and contact items are available in Rounds 6-7; frequency of religious practice in Round 6; importance of religion in Round 5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Several natural channels can be ruled out directly. *Denominational switching*: Table 9 shows that formative-year drought does not push Christians toward charismatic or Pentecostal denominations — the fast-growing prosperity-oriented movements — with point estimates that, if anything, lean toward mainline churches. *Schooling*: exposure at ages 12–17 has no effect on the level of education attained or on whether primary or secondary schooling is the highest level reached, in the full sample or among stayers (Appendix B.5), so the membership response does not run through disrupted schooling. *Material conditions*: formative-year exposure does not predict adult deprivation in food, water, medical care, fuel, or cash (Appendix Table B23), indicating that the membership response is not a reflection of persistently worse material circumstances.

Table 9: Denomination: No Shift Toward Charismatic Churches

	Charismatic (Christians)		Pentecostal		Protestant	
	Full	Stay	Full	Stay	Full	Stay
$Avg(Exp_{0-5})$	-0.007 (0.009)	-0.004 (0.012)	-0.002 (0.003)	0.003 (0.005)	-0.002 (0.006)	0.006 (0.009)
$Avg(Exp_{6-11})$	0.006 (0.008)	-0.003 (0.011)	-0.002 (0.003)	-0.001 (0.004)	0.004 (0.006)	0.008 (0.009)
$Avg(Exp_{12-17})$	-0.009 (0.009)	-0.018* (0.011)	-0.000 (0.004)	-0.002 (0.005)	-0.004 (0.005)	-0.002 (0.007)
$Avg(Exp_{18-23})$	-0.016* (0.009)	-0.023* (0.013)	-0.006** (0.003)	-0.008* (0.004)	-0.004 (0.006)	0.006 (0.009)
$Avg(Exp_{24-29})$	-0.001 (0.006)	0.001 (0.009)	-0.006** (0.003)	-0.006 (0.005)	-0.005 (0.005)	-0.014** (0.007)
$Avg(Exp_{30+})$	-0.012 (0.009)	-0.017 (0.013)	-0.002 (0.003)	-0.008 (0.005)	0.000 (0.006)	0.008 (0.010)
L x Y FE	Y	Y	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y	Y	Y
p(all brackets=0)	0.384	0.435	0.054	0.203	0.802	0.243
Observations	62,544	33,413	145,979	70,224	145,979	70,224
Num Localities	435	360	599	462	599	462
Num Countries	32	25	37	30	37	30
Adj R^2	0.201	0.186	0.100	0.095	0.305	0.299
Outcome Mean	0.294	0.282	0.056	0.064	0.370	0.407
Outcome SD	0.455	0.450	0.229	0.244	0.483	0.491

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. “Stayers” restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. “Charismatic” groups Pentecostal, Evangelical, Apostolic, Zionist, and African Independent denominations against mainline denominations, among Christians whose denomination is classifiable; unclassifiable responses are excluded. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Nor does the effect run through the timing of formal rites of passage. Matching each ethnic-religious group to an ethnographic coding of its rite of passage and its age window, the teenage drought effect is no larger for groups whose rite falls in the teenage years than for groups whose rite falls earlier or is absent (interaction -0.002 , s.e. 0.009 ; Appendix B.15). The adolescent imprint reflects the general formative role of adolescence — the socialization that rites of passage themselves institutionalize — rather than participation in any particular ceremony.

7.1. Short-Run Responses: Coping in Real Time

The results so far concern exposure years or decades before the survey. A natural question is whether membership also responds to drought as it happens — and whether any such response shows the same age profile as the durable imprint. To answer it, I construct a birth-cohort pseudo-panel: single-year-of-birth cells within each district are aggregated into five-year birth cohorts and tracked across the three survey rounds, so that cell fixed effects identify the response of the *same cohort's* membership share to recent drought, with the cell's gender and religious composition held constant (Appendix B.16).

Within cohort, each additional month of severe drought in the year before the survey raises the membership share by 0.8 percentage points ($p < 0.01$) — about 1.9 points for a one-standard-deviation shock of 2.4 severe months, comparable in magnitude to the formative-years effect. The estimate is unchanged under every cohort definition examined — band widths of three to ten years at all starting points (Appendix Table B20). But this short-run response differs from the durable one in every dimension that defines the latter. First, it is completely flat in age: the interaction with the cohort’s current age is 0.000 per decade (s.e. 0.001), and every current-age band from 18–27 to 58 and above responds by a statistically indistinguishable amount — adults of all ages turn to religious groups while sustained drought is under way. Second, it shows no gender difference ($p = 0.82$), where the durable imprint tilts male. Third, the evidence on its hardship channel is only suggestive: the share of the cell reporting food shortages rises with contemporaneous severe drought but imprecisely, employment does not move, and the response appears for *sustained* severe drought — the count of severe-drought months — but not for the annual-minimum measure that identifies the long-run effect (Appendix Table B21), consistent with duration of hardship, rather than a single dry month, driving contemporaneous coping. Taken together, the short-run evidence describes a temporary, age-neutral coping response — and it sharpens the interpretation of the main results: what distinguishes adolescent exposure is not that adolescents respond and others do not, but that only the attachments formed in the formative years persist after the drought has passed.

8. Robustness

8.1. Exposure Measurement

Table 10 re-estimates the age-bracket model for membership under five alternative exposure measures: the worst month of the wet season (April–September) and of the dry season, the annual *mean* SPEI — a symmetric measure that does not truncate at zero and aggregates by spatial averaging rather than the spatial minimum — and counts of moderate and severe drought months. The teenage coefficient is positive and significant at the 5% level under every measure. The symmetric annual-mean result is worth emphasizing: the effect does not depend on the drought-only construction of the baseline measure, and unusually wet formative years symmetrically reduce membership.

Table 10: Alternative Exposure Measures

Dependent Variable: Member						
	Annual min	Wet season	Dry season	Annual mean	Mod. droughts	Sev. droughts
<i>Avg(Exp₀₋₅)</i>	-0.003 (0.008)	-0.007 (0.010)	-0.009 (0.008)	0.002 (0.012)	-0.001 (0.002)	0.006** (0.003)
<i>Avg(Exp₆₋₁₁)</i>	0.007 (0.007)	0.005 (0.008)	0.002 (0.008)	0.007 (0.011)	0.000 (0.002)	0.003 (0.003)
<i>Avg(Exp₁₂₋₁₇)</i>	0.017** (0.006)	0.016** (0.008)	0.015** (0.006)	0.021** (0.010)	0.003** (0.001)	0.005** (0.002)
<i>Avg(Exp₁₈₋₂₃)</i>	0.011* (0.006)	-0.004 (0.008)	0.005 (0.007)	0.001 (0.011)	0.000 (0.002)	0.004* (0.002)
<i>Avg(Exp₂₄₋₂₉)</i>	0.009 (0.007)	0.001 (0.008)	0.006 (0.007)	0.009 (0.009)	0.001 (0.001)	0.004** (0.002)
<i>Avg(Exp₃₀₊)</i>	-0.007 (0.007)	-0.027** (0.012)	-0.015 (0.012)	-0.026 (0.017)	-0.003 (0.003)	-0.002 (0.003)
p(all brackets=0)	0.021	0.034	0.047	0.091	0.058	0.008
Observations	145,979	145,979	145,979	145,979	145,979	145,979
Adj <i>R</i> ²	0.187	0.187	0.187	0.187	0.187	0.187
Outcome Mean	0.465	0.465	0.465	0.465	0.465	0.465

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. Controlling for gender. Column heads name the exposure measure used to construct all brackets; the count measures (moderate/severe drought months) are in months per year, so coefficients are not directly comparable in magnitude to the SPEI-based columns. The corresponding estimates for *Religious* are small and mostly insignificant (Appendix B.6). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8.2. Inference

Because the SPEI is spatially interpolated, shocks are correlated across localities. Table 11 reports the headline bracket model under four inference schemes: clustering by country-year (baseline), by locality (ADM1), two-way clustering by both, and Conley (1999) spatial HAC standard errors with a 500-kilometer Bartlett kernel. The teenage standard error is essentially unchanged across all four, and the joint significance of the brackets is strongest under the spatial HAC correction.

Table 11: Alternative Inference

Dependent Variable: Member				
	Country-year	ADM1	Two-way	Conley 500km
<i>Avg(Exp₀₋₅)</i>	-0.003 (0.008)	-0.003 (0.007)	-0.003 (0.008)	-0.003 (0.007)
<i>Avg(Exp₆₋₁₁)</i>	0.007 (0.007)	0.007 (0.007)	0.007 (0.008)	0.007 (0.007)
<i>Avg(Exp₁₂₋₁₇)</i>	0.017** (0.006)	0.017** (0.007)	0.017** (0.007)	0.017** (0.007)
<i>Avg(Exp₁₈₋₂₃)</i>	0.011* (0.006)	0.011* (0.006)	0.011 (0.006)	0.011* (0.006)
<i>Avg(Exp₂₄₋₂₉)</i>	0.009 (0.007)	0.009 (0.007)	0.009 (0.007)	0.009 (0.007)
<i>Avg(Exp₃₀₊)</i>	-0.007 (0.007)	-0.007 (0.006)	-0.007 (0.007)	-0.007 (0.007)
p(all brackets=0)	0.021	0.011	0.027	0.001
Clusters	111	599	111	
Observations	145,979	145,979	145,979	145,594

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. The Conley column is estimated on the geocoded subsample (145,594 of 145,979 respondents) after partialling out the fixed effects; by the Frisch-Waugh theorem the coefficients are identical to the baseline and only the standard errors differ. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8.3. Locality-Specific Cohort Trends and a Placebo Test

The fixed effects in Equation (1) absorb cohort shocks common to a country and all time-varying differences across localities, but not differences in how successive cohorts evolve *within* a locality. Schooling provides a sharp test of whether such trends contaminate the age-bracket estimates: educational attainment is complete by the early twenties, so drought exposure at ages 24 and above cannot affect it, and any coefficient on those brackets must reflect residual cohort variation rather than a causal effect. Table 12 shows that the baseline specification fails this placebo for one outcome. Exposure at ages 24–29 and 30 and above predicts completing only primary school, with coefficients of -0.023 and -0.025 ($p < 0.01$) and a joint test of the two late brackets that rejects at any conventional level. Adding a locality-specific linear trend in year of birth removes the artifact: the late coefficients fall to -0.012 and 0.006 , the joint test no longer rejects ($p = 0.103$), and the two other schooling outcomes pass the placebo in both specifications.

Table 12: Placebo Test: Drought Exposure After Schooling Age and Educational Attainment

	Dependent Variable:					
	Educ. level	Educ. level	Highest: primary	Highest: primary	Highest: secondary	Highest: secondary
$Avg(Exp_{0-5})$	-0.006 (0.015)	-0.010 (0.015)	-0.001 (0.008)	0.001 (0.009)	-0.010 (0.007)	-0.005 (0.008)
$Avg(Exp_{6-11})$	-0.008 (0.013)	-0.019 (0.013)	-0.002 (0.008)	-0.002 (0.009)	-0.008 (0.008)	-0.005 (0.007)
$Avg(Exp_{12-17})$	0.004 (0.015)	-0.005 (0.016)	0.009 (0.007)	0.012 (0.009)	-0.010 (0.007)	-0.007 (0.008)
$Avg(Exp_{18-23})$	0.021* (0.012)	0.010 (0.014)	-0.011* (0.007)	-0.002 (0.009)	0.003 (0.007)	0.003 (0.007)
$Avg(Exp_{24-29})$	0.006 (0.012)	-0.005 (0.014)	-0.023*** (0.007)	-0.012 (0.008)	0.002 (0.007)	0.001 (0.007)
$Avg(Exp_{30+})$	-0.003 (0.017)	-0.012 (0.023)	-0.025*** (0.009)	0.006 (0.014)	-0.001 (0.009)	0.000 (0.013)
L x Y FE	Y	Y	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y	Y	Y
Locality x cohort trend	None	Linear	None	Linear	None	Linear
p(all brackets=0)	0.572	0.591	0.000	0.186	0.309	0.822
p(ages 24+ brackets=0)	0.816	0.869	0.000	0.103	0.942	0.986
Observations	145,543	145,543	145,543	145,543	145,543	145,543
Outcome Mean	1.461	1.461	0.298	0.298	0.364	0.364

Notes: Standard errors clustered by Country-Year in parentheses. L represents Location, Y represents Year, C represents Country, and YOB represents Year of Birth. Controlling for gender. Educ. level is the highest level attained (0 none, 1 primary, 2 secondary, 3 post-secondary); Highest: primary and Highest: secondary are indicators for that level being the highest attained. "Linear" columns add a locality-specific linear trend in year of birth (locality x year of birth). Because schooling is completed by the early twenties, the brackets for ages 24 and above are placebo regressors; "p(ages 24+ brackets=0)" is the joint test that they are zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The same trends sharpen the membership result rather than weakening it (Table 13). The teenage coefficient rises from 0.017 ($p < 0.05$) to 0.025 ($p < 0.01$), the active-membership coefficient is unchanged at 0.023 ($p < 0.01$), and religiousness remains jointly null. Under trends the profile is somewhat broader — the adjacent 6–11 and 18–23 brackets also reach significance at the 5% level — but its shape is preserved: the peak is at ages 12–17, the late brackets are jointly insignificant, and exposure after age 30 does nothing. A still more demanding specification, replacing the linear trend with locality x decade-of-birth fixed effects, leaves too little within-locality cohort variation to be informative: standard errors roughly double and the membership coefficient becomes imprecise (0.008 , s.e. 0.012), with a confidence interval that comfortably contains the baseline

estimate (Appendix Table B10). Among homeland stayers, where exposure is best measured, the teenage coefficient is 0.030 at baseline, 0.039 ($p < 0.01$) with linear trends, and 0.022 (s.e. 0.016) with decade fixed effects.

Table 13: Locality-Specific Cohort Trends

	Dependent Variable:					
	Member	Member	Active	Active	Religious	Religious
<i>Avg(Exp₀₋₅)</i>	-0.003 (0.008)	0.006 (0.007)	-0.006 (0.006)	-0.003 (0.007)	-0.001 (0.003)	-0.002 (0.003)
<i>Avg(Exp₆₋₁₁)</i>	0.007 (0.007)	0.015** (0.007)	0.004 (0.007)	0.006 (0.008)	-0.000 (0.003)	-0.001 (0.003)
<i>Avg(Exp₁₂₋₁₇)</i>	0.017** (0.006)	0.025*** (0.007)	0.022*** (0.006)	0.023*** (0.007)	-0.003 (0.003)	-0.004 (0.003)
<i>Avg(Exp₁₈₋₂₃)</i>	0.011* (0.006)	0.017** (0.008)	0.013** (0.006)	0.013* (0.007)	0.002 (0.003)	0.001 (0.003)
<i>Avg(Exp₂₄₋₂₉)</i>	0.009 (0.007)	0.015* (0.008)	0.007 (0.007)	0.007 (0.008)	0.000 (0.002)	-0.001 (0.003)
<i>Avg(Exp₃₀₊)</i>	-0.007 (0.007)	-0.003 (0.013)	-0.002 (0.007)	-0.003 (0.014)	-0.003 (0.003)	-0.010** (0.004)
L x Y FE	Y	Y	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y	Y	Y
Locality × cohort trend	None	Linear	None	Linear	None	Linear
p(all brackets=0)	0.021	0.003	0.000	0.004	0.679	0.138
p(ages 24+ brackets=0)	0.264	0.156	0.493	0.562	0.623	0.081
Observations	145,979	145,979	145,979	145,979	145,979	145,979
Outcome Mean	0.465	0.465	0.307	0.307	0.965	0.965

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. *Member* takes a value of 1 if the individual reports being a member of a religious group; *Active* if an active member or leader; *Religious* if the individual reports having a religion. “Linear” columns add a locality-specific linear trend in year of birth (locality × year of birth). “p(ages 24+ brackets= 0)” is the joint test that the two post-schooling brackets are zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8.4. Adaptation and Development

Two moderators speak to interpretation, and both are fixed before or at the time of exposure — unlike splits on adult occupation or wealth, which are themselves potential outcomes of the shock. First, interacting exposure with the locality’s long-run drought frequency (the share of years 1950–2010 in moderate drought) tests for adaptation in the sense of Dell et al. (2014): a formative-year drought could matter less where droughts are routine. It does not — the interaction is a precise zero ($p = 0.84$) — indicating little adaptation of the religious response to the local climate distribution. Second, interacting exposure with the country’s GDP per capita at the time the respondent was an adolescent, the effect is concentrated in poorer countries: the teenage coefficient is 0.025 ($p < 0.01$) below the sample median of GDP per capita and zero above it, though the continuous interaction is imprecise. This gradient matches the pattern documented for climate and conflict (Burke et al., 2024) and is what an informal-insurance reading predicts: where formal institutions and markets can smooth shocks, the congregation is not needed.

Table 14: Pre-Determined Moderators: Drought Frequency and Development

	Local drought frequency		GDP p.c. at exposure	
	Rare drought	Common drought	Poorer ctry	Richer ctry
$Avg(Exp_{0-5})$	-0.002 (0.011)	-0.014 (0.012)	0.002 (0.009)	-0.012 (0.013)
$Avg(Exp_{6-11})$	-0.002 (0.012)	0.019* (0.011)	0.000 (0.011)	0.009 (0.010)
$Avg(Exp_{12-17})$	0.013 (0.011)	0.019** (0.009)	0.025*** (0.009)	-0.000 (0.010)
$Avg(Exp_{18-23})$	0.022** (0.010)	0.011 (0.009)	0.009 (0.008)	0.009 (0.010)
$Avg(Exp_{24-29})$	0.003 (0.012)	0.013 (0.010)	0.020* (0.011)	-0.005 (0.009)
$Avg(Exp_{30+})$	0.002 (0.014)	0.001 (0.011)	-0.003 (0.011)	0.003 (0.011)
teen x moderator (z)	0.001	0.001	-0.007	-0.007
p(teen x moderator)	0.843	0.843	0.261	0.261
p(all brackets=0)	0.238	0.078	0.119	0.514
Observations	76,159	68,974	72,909	72,923
Num Localities	348	251	442	492
Num Countries	36	28	30	29
Adj R^2	0.200	0.174	0.163	0.200
Outcome Mean	0.451	0.479	0.507	0.423

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Columns split the sample at the main-sample median of each moderator; “teen x moderator” and its p -value come from a pooled model interacting every bracket with the standardized continuous moderator. Local drought frequency is the ADM1 share of years 1950–2010 with annual-minimum SPEI below -1 ; GDP per capita is the Maddison Project value for the respondent’s country at age 14. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9. Conclusion

This paper shows that droughts experienced during the formative teenage years durably raise participation in religious institutions across 37 African countries. The effect is precisely timed: it is concentrated in exposure at ages 12–19, fades through the early twenties, and is absent for exposure in early childhood or later adulthood, a profile that is robust across every bracketing scheme, exposure measure, and inference method examined. It operates on the margin of belonging rather than belief: religious affiliation, stated religiosity, denominational choice, and trust in religious leaders do not respond, while membership, active membership, and contact with religious leaders do.

The identification survives the central threat to designs of this kind. Using two independent proxies for having grown up in place — residence in the ethnic homeland and membership in the local ethnic majority — the effect is present and precisely estimated only among non-migrants, formative-year drought does not predict who stays, and homeland-based exposure behaves exactly as a coarse proxy for the lived environment should. The estimate also survives locality-specific cohort trends — which remove the one placebo failure in the design — as well as

randomization inference and leave-one-out exercises by country and by decade of adolescence. The response is religion-specific in adolescence, extending to secular community groups only at the threshold of adulthood and more weakly; it does not run through schooling, denominational switching, or persistent material deprivation; and it is concentrated in poorer countries while showing no attenuation in drought-prone places. A birth-cohort panel adds the short-run counterpart: while sustained severe drought is under way, the same cohort's membership rises temporarily at every adult age — underscoring that the durable imprint is specific to exposure in the formative years, not to contemporaneous hardship.

Together these results identify a demand-side channel through which climate volatility builds durable religious institutions: adverse shocks in the years when social attachments form convert into lifelong membership in the organizations best placed to insure against the next shock. Given that exposure to climatic volatility in Africa is projected to rise, and that the effect shows no sign of adaptation, the formative-years channel documented here suggests that climate change may reinforce the institutional centrality of religion precisely where formal insurance remains thin — with implications for how norms, social capital, and development interact over the coming decades.

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ONLINE APPENDIX FOR

**The Roots of Faith:
Early Life Droughts and Religious Commitment in Africa**

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8 October 2026

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Appendix A. Data Appendix

A.1. Afrobarometer Survey Overview

The information on individuals' demographic characteristics, location, and religious expression for this paper comes from the Afrobarometer survey. Conducted periodically since 1999, the Afrobarometer is designed to capture public sentiments and experiences related to democracy, governance, and socioeconomic conditions of individuals aged 18 and above across the African continent. For this study, I harmonized Rounds 5, 6, and 7 of the Afrobarometer, covering the period circa 2011 - 2018. The final sample for the analysis includes 145,979 observations from 599 sub-national units spanning 37 African countries: Algeria, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Côte d'Ivoire, Egypt, Eswatini, Gabon, Gambia, Ghana, Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, South Africa, Sudan, Tanzania, Togo, Tunisia, Uganda, Zambia, Zimbabwe.

A.1.1. Variables and Harmonization Details

From the Afrobarometer, I use individuals' age, employment status, date of survey, location, sex, race, ethnicity, religion, and religious group membership status. For location, some boundaries changed during the period of these surveys. To account for this, I matched them according to the references of these administrative changes, mostly by aggregating them to the largest comparable area across time. The religion variable divides the sample into four categories: Muslim, Christian, Other, and None. While Round 7 includes this condensed version of the religion variable, it is unavailable in Round 6. To make them comparable, I use the religion mapping from the condensed and raw religion variables in Round 7 and apply it to the data from Round 6. The question regarding religious group membership status asks whether the individual is an official leader, an active member, an inactive member, or not a member of a religious group that meets outside of regular worship services. This question was not exclusively asked to those who report being religious; around 15% of individuals who report having no religion declared that they are a member of a religious group, either active or inactive. To combine the individual-level demographic and religion information with the climate data, which allows me to construct drought measurements for the analysis, I cleaned and harmonized the subnational division names in the Afrobarometer and used the geolocation of the surveys to match them with the shapefiles from the Global Administrative Areas, GADM (Version 4.1, released on July 16, 2022).

A.2. Drought Measurement

This appendix provides a detailed description of the methodology and data used to construct the drought measurements for this study.

A.2.1. The Global SPEI Database

To build my measures of drought exposure and intensity, I use The Global SPEI Database. This dataset offers comprehensive information on drought severity and duration worldwide, drawing from monthly precipitation and evapotranspiration data from the Climatic Research Unit of the University of East Anglia, to build a global gridded dataset of a multiscalar drought index — the standardized precipitation evapotranspiration index (SPEI)³, which

³Based on the Palmer drought severity index (UCAR-PDSI, 2.5°).

considers the joint effects of temperature and precipitation on droughts. Version 2.9 of the database covers the period from January 1901 to December 2022, providing a detailed record of meteorological conditions over more than a century. Following Vicente-Serrano et al. (2010), I use the 12-month Standardized Precipitation-Evapotranspiration Index (SPEI-12), to identify droughts from water deficits in a rolling 12-month period, allowing for the precise characterization of drought events and their temporal variability across the countries in the sample.

A.2.2. Drought Categories

The SPEI standardized scale defines the following categories of drought:

- Slight: $(0, -0.9)$
- Moderate: $[-0.9, -1.5)$
- Severe: $[-1.5, -2)$
- Extreme: $[-2, -\infty)$

A.2.3. Drought Variable Construction

For my initial regression analysis, exploring the link between overall drought exposure and religiosity, I combine the SPEI spatial information with the GADM shapefiles and the year-of-birth and location of individuals in the sample to construct different measures of drought exposure for these individuals based on their location, year of birth, and survey year: Avg(Drought Exposure). Avg(Drought Exposure) consists of the cumulative level of drought exposure by adding up the SPEI measure from the year in which the individual is born until the year of the interview⁴. I then divide by each individual's age to calculate the average exposure and intensity of exposure during their lifetime.

Additionally, to explore the life-cycle component of exposure to droughts and its relationship with religion, I construct the same cumulative measures for Drought Exposure averaged within six age brackets spanning ages 0–5, 6–11, 12–17, 18–23, 24–29, and 30 and above. These brackets are chosen to align with developmental life stages — in particular, the 12–17 bracket captures the adolescent window emphasized throughout this paper — and the results are robust to alternative bracket widths and start points.

⁴I ignore positive values of the variable, only focusing on the driest measurement of my monthly 12-month moving average drought measurement for each pixel of the map, to later take the average across all pixels for each location.

Appendix B. Additional Tables & Figures

B.1. Droughts and Religiousness: Exposure based on Ethnic Origins

Table B1: Impact of Droughts on Religiousness and Group Membership

	Dependent Variable:			
	Religious		Member	
	(1)	(2)	(3)	(4)
$Avg_o(Exp)$	0.012* (0.006)		-0.021* (0.011)	
$Avg(Exp_{0-5,o})$		0.004 (0.002)		0.001 (0.007)
$Avg(Exp_{6-11,o})$		0.002 (0.002)		-0.009 (0.007)
$Avg(Exp_{12-17,o})$		0.001 (0.002)		-0.000 (0.006)
$Avg(Exp_{18-23,o})$		0.005** (0.002)		0.004 (0.005)
$Avg(Exp_{24-29,o})$		-0.001 (0.003)		-0.006 (0.006)
$Avg(Exp_{30+})$		-0.000 (0.002)		-0.012** (0.005)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Observations	145,979	145,979	145,979	145,979
Num Localities	599	599	599	599
Num Countries	37	37	37	37
Adj R^2	0.084	0.084	0.187	0.187
Outcome Mean	0.965	0.965	0.465	0.465
Outcome SD	0.184	0.184	0.499	0.499

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Drought exposure is computed at the Murdock ethnic-homeland polygon for matched respondents and backfilled with current-location exposure for unmatched respondents; Appendix Table B3 reports the matched-only version with ethnic-homeland fixed effects. Restricted to the sample of the main results. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2. Ethnic-Homeland Exposure: Matched Sample with Homeland Fixed Effects

Table B2: Impact of Droughts: Ethnic-Homeland Exposure, Matched Respondents Only

	Dependent Variable:			
	Religious		Member	
	(1)	(2)	(3)	(4)
$Avg_o(Exp)$	0.013* (0.007)		-0.022* (0.012)	
$Avg(Exp_{0-5,o})$		0.004 (0.003)		0.004 (0.008)
$Avg(Exp_{6-11,o})$		0.001 (0.003)		-0.014* (0.007)
$Avg(Exp_{12-17,o})$		0.001 (0.003)		0.001 (0.007)
$Avg(Exp_{18-23,o})$		0.007*** (0.003)		0.002 (0.006)
$Avg(Exp_{24-29,o})$		0.000 (0.003)		-0.005 (0.007)
$Avg(Exp_{30+})$		-0.001 (0.003)		-0.009 (0.006)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Observations	111,286	111,286	111,286	111,286
Num Localities	501	501	501	501
Num Countries	31	31	31	31
Adj R^2	0.076	0.076	0.154	0.154
Outcome Mean	0.964	0.964	0.503	0.503
Outcome SD	0.185	0.185	0.500	0.500

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Sample restricted to respondents with a Murdock ethnic-homeland match; exposure is computed at the homeland polygon with no backfilling. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B3: Ethnic-Homeland Exposure with Homeland Fixed Effects

	Dependent Variable:			
	Religious		Member	
	(1)	(2)	(3)	(4)
$Avg_o(Exp)$	0.003 (0.009)		-0.048 (0.034)	
$Avg(Exp_{0-5,o})$		0.002 (0.003)		-0.001 (0.009)
$Avg(Exp_{6-11,o})$		0.000 (0.002)		-0.017* (0.009)
$Avg(Exp_{12-17,o})$		0.000 (0.003)		0.003 (0.008)
$Avg(Exp_{18-23,o})$		0.007*** (0.003)		0.005 (0.007)
$Avg(Exp_{24-29,o})$		0.000 (0.003)		-0.002 (0.008)
$Avg(Exp_{30+})$		0.001 (0.003)		-0.007 (0.006)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Observations	111,285	111,285	111,285	111,285
Num Localities	501	501	501	501
Num Countries	31	31	31	31
Num Homelands	303	303	303	303
Adj R^2	0.082	0.082	0.156	0.156
Outcome Mean	0.964	0.964	0.503	0.503
Outcome SD	0.185	0.185	0.500	0.500

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. As the previous table, adding fixed effects for the matched Murdock homeland, so that homeland exposure is identified from cohort variation within an ethnic group. Without these fixed effects, homeland exposure varies primarily across ethnic groups and confounds drought with ethnic composition. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3. Migration: Additional Results

Table B4: Stayer Definitions: Threshold Robustness

	Dependent Variable: Member						
	Home 25km	Home 50km	Home 100km	Home 50km same cties	Share .3	Share .5	Share .7
$Avg(Exp_{0-5})$	0.009 (0.012)	0.005 (0.011)	0.004 (0.010)	0.004 (0.011)	0.003 (0.009)	0.007 (0.010)	0.008 (0.010)
$Avg(Exp_{6-11})$	0.005 (0.012)	0.008 (0.011)	0.008 (0.010)	0.007 (0.011)	0.014 (0.009)	0.011 (0.010)	0.017 (0.011)
$Avg(Exp_{12-17})$	0.032*** (0.010)	0.030*** (0.010)	0.028*** (0.008)	0.028*** (0.010)	0.027*** (0.008)	0.025*** (0.009)	0.027*** (0.009)
$Avg(Exp_{18-23})$	0.018* (0.010)	0.017* (0.009)	0.010 (0.008)	0.017* (0.009)	0.012* (0.007)	0.006 (0.007)	0.004 (0.008)
$Avg(Exp_{24-29})$	0.021** (0.011)	0.017* (0.010)	0.012 (0.009)	0.019* (0.009)	0.016* (0.008)	0.016* (0.008)	0.015 (0.009)
$Avg(Exp_{30+})$	0.002 (0.012)	-0.003 (0.011)	-0.004 (0.010)	-0.004 (0.011)	-0.002 (0.009)	0.000 (0.009)	0.006 (0.012)
p(all brackets=0)	0.010	0.018	0.022	0.021	0.010	0.070	0.063
Observations	61,942	70,224	82,024	68,657	99,582	85,916	65,439
Num Localities	453	462	479	439	575	570	525
Num Countries	30	30	31	30	36	36	35
Adj R^2	0.157	0.158	0.155	0.158	0.178	0.183	0.187
Outcome Mean	0.518	0.519	0.515	0.519	0.481	0.480	0.477
Outcome SD	0.500	0.500	0.500	0.500	0.500	0.500	0.499

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. Controlling for gender. Each column estimates the age-bracket model on a different stayer definition: within 25/50/100 km of the ethnic-homeland polygon, the 50 km definition restricted to countries with both stayers and movers, and own-ethnicity locality shares of at least 0.3/0.5/0.7. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4. Stayers versus Movers: Religious Affiliation

Table B5: Stayers versus Movers: Religious Affiliation

	Ethnic Homeland		Locality Ethnic Share	
	In homeland	Outside	Majority	Minority
$Avg(Exp_{0-5})$	0.002 (0.004)	-0.004 (0.004)	0.001 (0.003)	-0.010* (0.005)
$Avg(Exp_{6-11})$	0.001 (0.003)	0.001 (0.006)	0.003 (0.003)	-0.003 (0.005)
$Avg(Exp_{12-17})$	-0.001 (0.003)	-0.006 (0.005)	-0.003 (0.003)	-0.003 (0.006)
$Avg(Exp_{18-23})$	0.008** (0.003)	0.005 (0.005)	0.004 (0.003)	-0.004 (0.008)
$Avg(Exp_{24-29})$	0.003 (0.004)	0.003 (0.005)	-0.001 (0.003)	0.006 (0.005)
$Avg(Exp_{30+})$	0.001 (0.003)	-0.002 (0.006)	-0.001 (0.004)	-0.006 (0.005)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
p(all brackets=0)	0.304	0.650	0.447	0.185
p(teen effect equal across split)	0.408	0.408	0.970	0.970
Observations	70,224	40,620	85,916	51,526
Num Localities	462	458	570	512
Num Countries	30	31	36	34
Adj R^2	0.097	0.056	0.099	0.070
Outcome Mean	0.968	0.960	0.966	0.959
Outcome SD	0.177	0.197	0.180	0.199

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. Controlling for gender. Dependent variable: *Religious*. Sample splits as in Table 4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B6: Does Formative-Year Drought Predict Urban Residence?

	Dependent Variable: Urban	
	Urban (full)	Urban (stayers)
$Avg(Exp_{0-5})$	0.000 (0.006)	0.001 (0.009)
$Avg(Exp_{6-11})$	-0.001 (0.005)	0.014* (0.008)
$Avg(Exp_{12-17})$	0.003 (0.006)	0.011 (0.007)
$Avg(Exp_{18-23})$	-0.003 (0.006)	0.003 (0.009)
$Avg(Exp_{24-29})$	-0.000 (0.007)	-0.003 (0.010)
$Avg(Exp_{30+})$	-0.010 (0.007)	-0.006 (0.009)
L x Y FE	Y	Y
C x Y x YOB FE	Y	Y
p(all brackets=0)	0.762	0.354
Observations	145,979	70,224
Num Localities	599	462
Num Countries	37	30
Adj R^2	0.395	0.328
Outcome Mean	0.407	0.321
Outcome SD	0.491	0.467

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender (urban residence is excluded from the controls). Each column regresses an indicator for urban residence at the time of the survey on the full set of drought-exposure brackets; column (2) restricts the sample to homeland stayers, for whom a move to town within the homeland area is the remaining margin of relocation.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5. Schooling

Table B7: Impact of Droughts on Educational Attainment

	Education Level (0–3)		Highest Level: Primary		Highest Level: Secondary	
	Full	Stay	Full	Stay	Full	Stay
$Avg(Exp_{0-5})$	−0.006 (0.015)	0.008 (0.017)	−0.001 (0.008)	−0.008 (0.012)	−0.010 (0.007)	−0.000 (0.010)
$Avg(Exp_{6-11})$	−0.008 (0.013)	0.010 (0.017)	−0.002 (0.008)	−0.014 (0.013)	−0.008 (0.008)	0.008 (0.012)
$Avg(Exp_{12-17})$	0.004 (0.015)	0.026 (0.020)	0.009 (0.007)	−0.005 (0.010)	−0.010 (0.007)	0.015 (0.009)
$Avg(Exp_{18-23})$	0.021* (0.012)	0.050*** (0.016)	−0.011* (0.007)	−0.010 (0.009)	0.003 (0.007)	0.018* (0.010)
$Avg(Exp_{24-29})$	0.006 (0.012)	0.001 (0.016)	−0.023*** (0.007)	−0.018* (0.010)	0.002 (0.007)	0.007 (0.009)
$Avg(Exp_{30+})$	−0.003 (0.017)	0.019 (0.022)	−0.025*** (0.009)	−0.033*** (0.011)	−0.001 (0.009)	0.012 (0.011)
L x Y FE	Y	Y	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y	Y	Y
p(all brackets=0)	0.572	0.120	0.000	0.067	0.309	0.324
Observations	145,543	70,036	145,543	70,036	145,543	70,036
Num Localities	599	462	599	462	599	462
Num Countries	37	30	37	30	37	30
Adj R^2	0.344	0.337	0.166	0.173	0.173	0.177
Outcome Mean	1.461	1.341	0.298	0.331	0.364	0.339
Outcome SD	0.962	0.940	0.457	0.471	0.481	0.473

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. Controlling for gender. “Stayers” restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. *Education Level* is the highest level attained (0 none, 1 primary, 2 secondary, 3 post-secondary); *Highest Level: Primary* and *Highest Level: Secondary* are indicators for that level being the highest attained. Coefficients on exposure brackets after schooling age cannot be causal; the significant late-bracket coefficients for primary completion disappear once locality-specific cohort trends are added (Section 8.3, Table 12). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.6. Robustness: Additional Tables

Table B8: Alternative Exposure Measures: Religious Affiliation

Dependent Variable: Religious						
	Annual min	Wet season	Dry season	Annual mean	Mod. droughts	Sev. droughts
$Avg(Exp_{0-5})$	-0.001 (0.003)	-0.002 (0.003)	-0.001 (0.003)	-0.000 (0.005)	0.000 (0.001)	0.002 (0.001)
$Avg(Exp_{6-11})$	-0.000 (0.003)	-0.001 (0.003)	-0.001 (0.003)	-0.001 (0.003)	-0.000 (0.001)	0.001 (0.001)
$Avg(Exp_{12-17})$	-0.003 (0.003)	-0.008** (0.004)	-0.006** (0.003)	-0.007 (0.005)	-0.001* (0.001)	-0.001 (0.001)
$Avg(Exp_{18-23})$	0.002 (0.003)	-0.001 (0.004)	-0.000 (0.003)	-0.001 (0.004)	-0.001 (0.001)	0.000 (0.001)
$Avg(Exp_{24-29})$	0.000 (0.002)	-0.002 (0.004)	-0.003 (0.003)	-0.003 (0.004)	0.000 (0.001)	0.000 (0.001)
$Avg(Exp_{30+})$	-0.003 (0.003)	-0.012* (0.006)	-0.010** (0.005)	-0.017** (0.007)	-0.003** (0.001)	-0.004*** (0.001)
p(all brackets=0)	0.679	0.283	0.189	0.151	0.102	0.008
Observations	145,979	145,979	145,979	145,979	145,979	145,979
Adj R^2	0.084	0.084	0.084	0.084	0.084	0.084
Outcome Mean	0.965	0.965	0.965	0.965	0.965	0.965

Notes: Standard errors clustered by Country-Year in parentheses. L represents Location, Y represents Year, C represents Country, and YOB represents Year of Birth. Controlling for gender. As Table 10, for Religious. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B9: Pre-Determined Moderators: Religious Affiliation

	Local drought frequency		GDP p.c. at exposure	
	Rare drought	Common drought	Poorer ctry	Richer ctry
$Avg(Exp_{0-5})$	-0.003 (0.004)	-0.002 (0.005)	-0.000 (0.003)	-0.001 (0.005)
$Avg(Exp_{6-11})$	-0.005 (0.004)	0.004 (0.004)	0.004 (0.003)	-0.002 (0.004)
$Avg(Exp_{12-17})$	-0.010* (0.005)	0.001 (0.005)	0.001 (0.004)	-0.007 (0.005)
$Avg(Exp_{18-23})$	-0.004 (0.006)	0.008** (0.004)	0.004 (0.003)	0.006 (0.004)
$Avg(Exp_{24-29})$	-0.002 (0.006)	0.004 (0.004)	0.003 (0.003)	0.002 (0.004)
$Avg(Exp_{30+})$	-0.003 (0.007)	0.007 (0.006)	0.002 (0.004)	-0.002 (0.005)
teen x moderator (z)	0.003	0.003	-0.002	-0.002
p(teen x moderator)	0.127	0.127	0.469	0.469
p(all brackets=0)	0.455	0.413	0.756	0.409
Observations	76,159	68,974	72,909	72,923
Num Localities	348	251	442	492
Num Countries	36	28	30	29
Adj R^2	0.087	0.080	0.070	0.092
Outcome Mean	0.965	0.965	0.974	0.957

Notes: Standard errors clustered by Country-Year in parentheses. L represents Location, Y represents Year, C represents Country, and YOB represents Year of Birth. Controlling for gender. As Table 14, for Religious. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.7. Validation: Stricter Trends, Randomization Inference, Stability, and Selection

This appendix collects the validation exercises referred to in Sections 4 and 8.3. Table B10 replaces the locality-specific linear cohort trend with locality $\times$ decade-of-birth fixed effects. Table B11 and Figure B1 report randomization inference for the teenage coefficient. Figures B2 and B3 and Table B12 examine the stability of the estimate across countries, decades, and cohort ranges. Table B13 tests for selection into the sample, and Table B14 reports ethnic-homeland exposure among homeland stayers.

Table B10: Locality $\times$ Decade-of-Birth Fixed Effects

	Full sample		Homeland stayers		
	Member	Highest: primary	Member (stay)	Member (stay)	Member (stay)
$Avg(Exp_{0-5})$	0.004 (0.010)	0.011 (0.013)	0.005 (0.011)	0.008 (0.011)	0.004 (0.017)
$Avg(Exp_{6-11})$	0.011 (0.012)	-0.006 (0.013)	0.008 (0.011)	0.014 (0.010)	0.007 (0.019)
$Avg(Exp_{12-17})$	0.008 (0.012)	0.007 (0.012)	0.030*** (0.010)	0.039*** (0.010)	0.022 (0.016)
$Avg(Exp_{18-23})$	0.006 (0.010)	-0.008 (0.009)	0.017* (0.009)	0.027** (0.013)	0.021 (0.013)
$Avg(Exp_{24-29})$	0.013 (0.010)	-0.014 (0.009)	0.017* (0.010)	0.027** (0.011)	0.011 (0.014)
$Avg(Exp_{30+})$	-0.009 (0.015)	0.013 (0.012)	-0.003 (0.011)	0.007 (0.023)	-0.020 (0.019)
L $\times$ Y FE	Y	Y	Y	Y	Y
C $\times$ Y $\times$ YOB FE	Y	Y	Y	Y	Y
Locality $\times$ cohort trend	Decade FE	Decade FE	None	Linear	Decade FE
p(all brackets=0)	0.857	0.181	0.018	0.005	0.430
p(ages 24+ brackets=0)	0.298	0.051	0.200	0.049	0.366
Observations	145,578	145,145	70,224	70,224	69,817
Outcome Mean	0.465	0.298	0.519	0.519	0.519

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. "Decade FE" columns add locality $\times$ decade-of-birth fixed effects, so that only variation across single birth years within a locality-decade identifies the brackets; "Linear" adds a locality-specific linear trend in year of birth, as in Table 13. Columns (3)–(5) restrict the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. Standard errors under decade fixed effects are roughly twice the baseline, and the confidence intervals contain the baseline estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

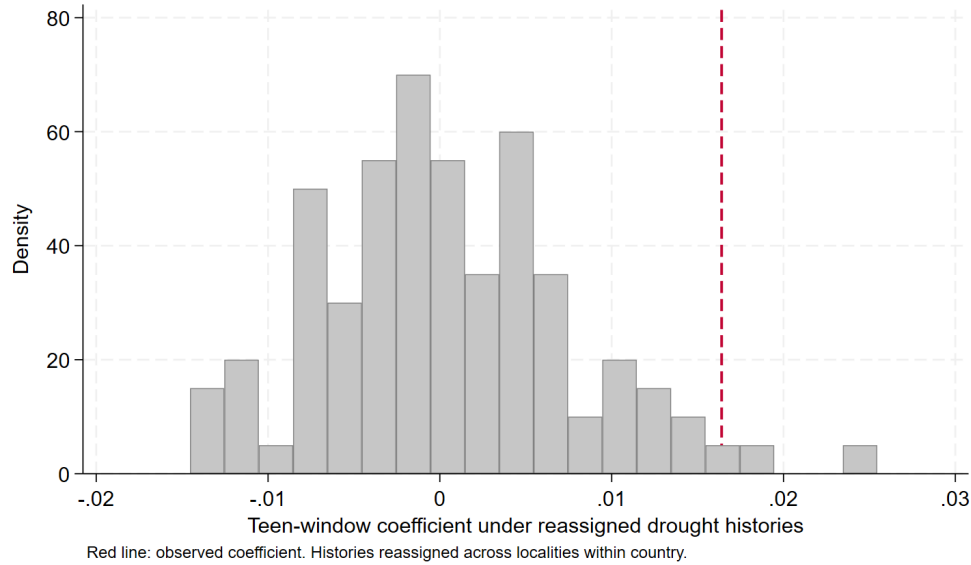
Randomization inference. Because drought is spatially and serially correlated, the effective number of independent drought histories is smaller than the number of locality-cohort cells, and model-based clustering may understate uncertainty. As a model-free check, I reassign each locality's *entire* drought history to a randomly chosen other locality in the same country (a random permutation of localities within country), rebuild every respondent's per-age exposure and age brackets from the reassigned history, and re-estimate the teenage coefficient. This preserves the within-country distribution of drought histories, their serial correlation, and the survey's cohort-location structure, and breaks only the link between a locality's actual history and its respondents. Table B11 and Figure B1 summarize 100 permutations: the observed coefficient lies in the upper tail of the placebo distribution, with a two-sided randomization p -value of 0.030, in line with the clustered inference.

Table B11: Randomization Inference for the Teenage-Window Coefficient

Observed teen coefficient	0.016
Clustered s.e.	0.006
Randomization-inference p (two-sided)	0.030
S.d. of placebo coefficients	0.0073
5th–95th pct. of placebo coefficients	-0.011 to 0.013
Permutations	100

Notes: Dependent variable: *Member*. Each permutation reassigns the full locality-by-year drought history across localities within a country, rebuilds per-age exposure and the six age brackets for every respondent, and re-estimates Equation (1). For the comparison to be like for like, the brackets are rebuilt from exposure at ages 0–44 for both the observed and the permuted estimates, which is why the observed coefficient (0.016) differs marginally from Table 2 (0.017). The randomization p -value is the share of permutations whose coefficient is at least as large in absolute value as the observed one.

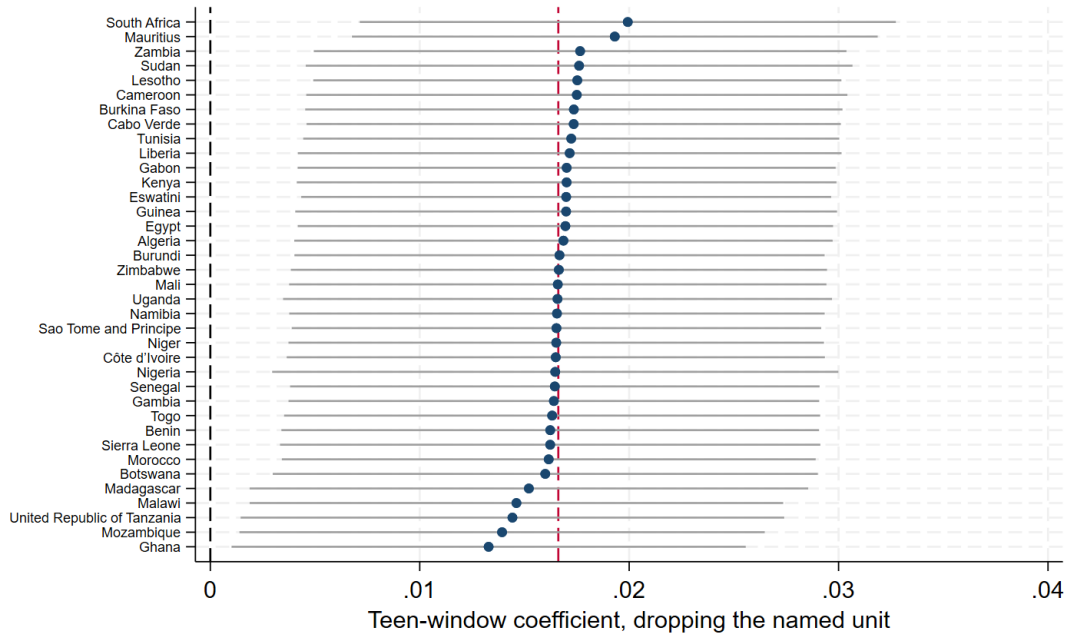
Figure B1: Placebo Distribution of the Teenage-Window Coefficient



Notes: Histogram of the teenage-window coefficient on *Member* across 100 permutations that reassign entire drought histories across localities within a country (Table B11). The dashed red line marks the observed coefficient.

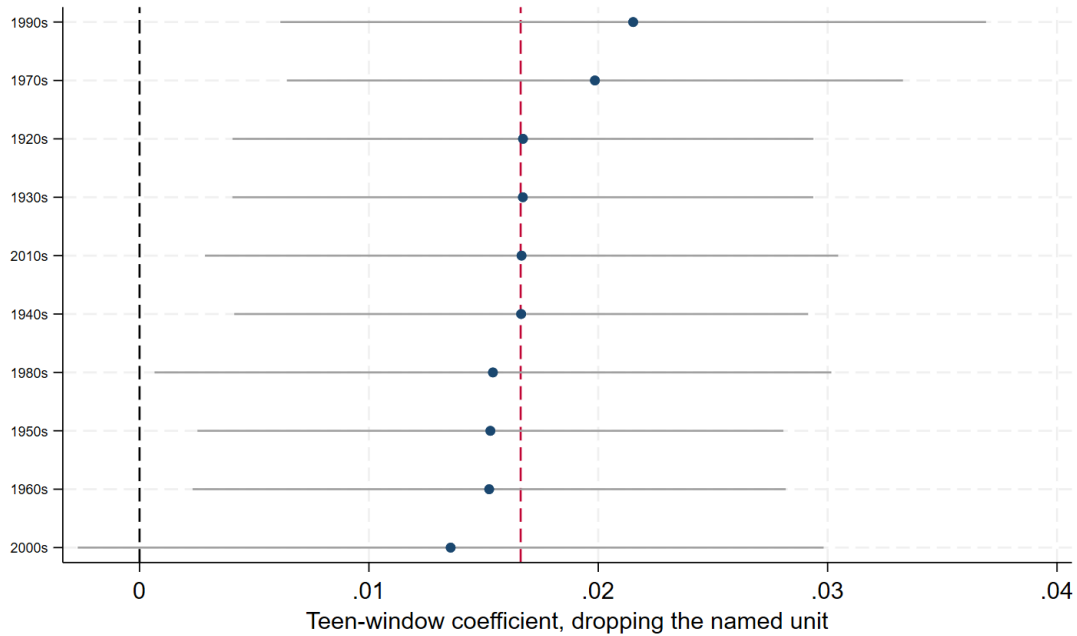
Stability. Figure B2 re-estimates the baseline age-bracket model dropping one country at a time; the teenage coefficient ranges from 0.013 (dropping Ghana) to 0.020 (dropping South Africa) around the full-sample value of 0.017, so no single country drives the result. Figure B3 drops instead all respondents whose teenage years fell in a given calendar decade; the coefficient ranges from 0.014 to 0.022, so the result is not carried by one drought episode. Table B12 restricts the sample to respondents old enough to have lived through every identifying window — ages 30, 36, and 48 and above — so that the age-bracket coefficients are identified from balanced cohort ranges rather than partly from the youngest cohorts. The teenage coefficient is 0.014, 0.016, and 0.033 ($p < 0.10$) in the three samples; precision falls with the sample, as expected, and the age profile is unchanged.

Figure B2: Leave-One-Country-Out Estimates



Notes: Each point is the teenage-window coefficient on *Member* from the baseline age-bracket model (Table 2, column (4)) re-estimated without the named country; bars are 95% confidence intervals, standard errors clustered by Country-Year. The dashed red line marks the full-sample estimate.

Figure B3: Leave-One-Decade-Out Estimates



Notes: Each point is the teenage-window coefficient on *Member* from the baseline age-bracket model re-estimated without the respondents whose teenage years (ages 12–17) fell in the named calendar decade; bars are 95% confidence intervals, standard errors clustered by Country-Year. The dashed red line marks the full-sample estimate.

Table B12: Balanced Cohort Ranges

	Dependent Variable: Member		
	Age ≥ 30	Age ≥ 36	Age ≥ 48
$Avg(Exp_{0-5})$	0.002 (0.010)	0.000 (0.012)	-0.002 (0.016)
$Avg(Exp_{6-11})$	0.011 (0.010)	0.019 (0.012)	0.012 (0.021)
$Avg(Exp_{12-17})$	0.014 (0.009)	0.016 (0.011)	0.033* (0.017)
$Avg(Exp_{18-23})$	0.009 (0.010)	0.020* (0.012)	0.017 (0.017)
$Avg(Exp_{24-29})$	0.006 (0.009)	0.009 (0.013)	0.021 (0.020)
$Avg(Exp_{30+})$	-0.010 (0.011)	-0.006 (0.013)	0.005 (0.017)
L x Y FE	Y	Y	Y
C x Y x YOB FE	Y	Y	Y
p(all brackets=0)	0.191	0.138	0.556
Observations	91,279	66,023	33,159
Adj R^2	0.193	0.196	0.191
Outcome Mean	0.477	0.482	0.484

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Each column restricts the main sample to respondents at or above the stated age at the survey, so that every respondent has lived through all brackets up to the stated age. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Selection into the sample. The Afrobarometer samples the resident adult population, so cohorts that lost members to drought-induced mortality or emigration would be under-represented in their locality. The survey offers one handle on this: the number of respondents in each locality $\times$ birth-year $\times$ survey-round cell. Table B13 regresses the log and the level of cell size on the cell's exposure brackets under the main fixed effects, with and without locality-specific cohort trends. Teenage exposure does not predict cell size in any specification (coefficients of -0.007 in logs), and the brackets are jointly insignificant once trends are included; the only baseline coefficient of note, on exposure after age 30, changes sign under trends and so reflects cohort trends rather than selection.

Table B13: Selection Test: Does Formative-Year Drought Predict Cell Size?

	ln(cell size)	ln(cell size), trends	Cell size	Cell size, trends
$Avg(Exp_{0-5})$	0.030* (0.018)	0.016 (0.014)	0.081 (0.101)	-0.007 (0.063)
$Avg(Exp_{6-11})$	0.005 (0.016)	0.002 (0.017)	-0.012 (0.075)	-0.065 (0.052)
$Avg(Exp_{12-17})$	-0.007 (0.016)	-0.007 (0.015)	-0.004 (0.092)	-0.055 (0.050)
$Avg(Exp_{18-23})$	0.020 (0.016)	0.019 (0.015)	0.080 (0.093)	0.025 (0.050)
$Avg(Exp_{24-29})$	-0.006 (0.018)	-0.005 (0.017)	-0.048 (0.099)	-0.083 (0.061)
$Avg(Exp_{30+})$	0.054** (0.022)	-0.024 (0.026)	0.310*** (0.115)	-0.181* (0.097)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
p(all brackets=0)	0.025	0.427	0.005	0.223
Cells	52,312	52,312	52,312	52,312
Adj R^2	0.532	0.562	0.576	0.668
Outcome Mean	0.699	0.699	2.769	2.769

Notes: The unit of observation is a locality $\times$ year-of-birth $\times$ survey-round cell of the main estimation sample; the dependent variable is the number of respondents in the cell (in logs or levels), and the regressors are the cell's drought-exposure brackets. Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. "Trends" columns add a locality-specific linear trend in year of birth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Homeland exposure among stayers. Section 6 attributes the null on ethnic-homeland exposure in the full matched sample to the coarseness of the homeland polygon as a proxy for the childhood environment. If that reading is right, homeland exposure should recover an attenuated but positive effect among respondents who live in their homeland, where the polygon and the lived environment overlap, and nothing among those who live outside it. Table B14 confirms both predictions: among homeland stayers the homeland-based teenage coefficient is 0.013 (s.e. 0.009), or 0.014 with homeland fixed effects, against 0.030 ($p < 0.01$) for current-location exposure in the same sample; among movers it is -0.023 ($p < 0.10$).

Table B14: Ethnic-Homeland Exposure Among Homeland Stayers and Movers

	Dependent Variable: Member			
	Residence, stayers	Homeland, stayers	Homeland, stayers + homeland FE	Homeland, movers
$Avg(Exp_{0-5})$	0.005 (0.011)			
$Avg(Exp_{6-11})$	0.008 (0.011)			
$Avg(Exp_{12-17})$	0.030*** (0.010)			
$Avg(Exp_{18-23})$	0.017* (0.009)			
$Avg(Exp_{24-29})$	0.017* (0.010)			
$Avg(Exp_{30+})$	-0.003 (0.011)			
$Avg(Exp_{0-5,o})$		-0.000 (0.011)	0.001 (0.011)	0.006 (0.011)
$Avg(Exp_{6-11,o})$		-0.011 (0.011)	-0.010 (0.012)	-0.012 (0.010)
$Avg(Exp_{12-17,o})$		0.013 (0.009)	0.014 (0.009)	-0.023* (0.012)
$Avg(Exp_{18-23,o})$		0.006 (0.010)	0.010 (0.010)	-0.001 (0.010)
$Avg(Exp_{24-29,o})$		0.003 (0.009)	0.004 (0.010)	-0.006 (0.012)
$Avg(Exp_{30+,o})$		-0.007 (0.010)	-0.005 (0.008)	0.002 (0.009)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Homeland FE	N	N	Y	N
p(all brackets=0)	0.018	0.599	0.511	0.151
Observations	70,224	70,220	70,215	39,982
Adj R^2	0.158	0.158	0.160	0.148
Outcome Mean	0.519	0.519	0.519	0.476

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. $Avg(Exp_{a-b})$ is exposure measured at the respondent's current locality; $Avg(Exp_{a-b,o})$ is exposure measured at the matched Murdock ethnic-homeland polygon. "Stayers" are respondents living within 50 km of their homeland polygon, "movers" those living farther away. Column (3) adds ethnic-homeland fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.8. Heterogeneity by Adult Characteristics

Table B15 reports the teenage effect split by urban residence, agricultural occupation, and lived poverty. These characteristics are measured in adulthood and are therefore potential outcomes of formative-year exposure; the pre-determined moderators in Table 14 are the preferred heterogeneity results. No split is statistically distinguishable, though point estimates are larger for agricultural workers.

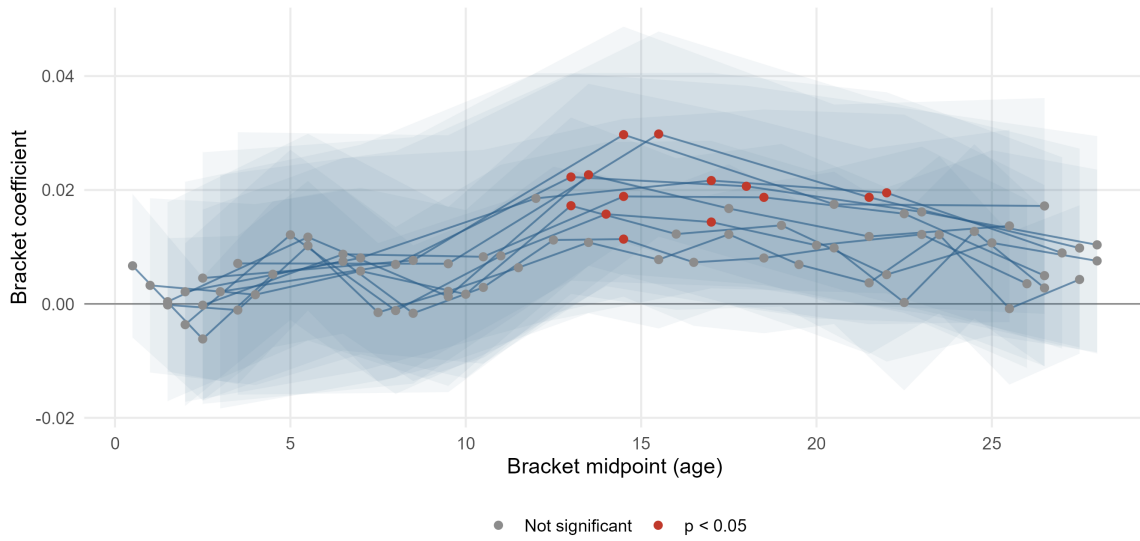
Table B15: Heterogeneity by Adult Characteristics (Post-Treatment; For Reference)

	Dependent Variable: Member					
	Rural	Urban	Non-agri	Agri	Less poor	Poorer
$Avg(Exp_{0-5})$	-0.011 (0.009)	0.014 (0.015)	0.000 (0.011)	-0.022 (0.018)	-0.014 (0.013)	0.005 (0.010)
$Avg(Exp_{6-11})$	0.013 (0.010)	0.003 (0.011)	0.013 (0.009)	0.003 (0.021)	0.015 (0.011)	-0.002 (0.010)
$Avg(Exp_{12-17})$	0.011 (0.009)	0.031** (0.015)	0.016* (0.008)	0.040** (0.016)	0.013 (0.011)	0.016** (0.008)
$Avg(Exp_{18-23})$	0.011 (0.007)	0.014 (0.010)	0.011 (0.007)	0.012 (0.019)	0.009 (0.009)	0.012 (0.008)
$Avg(Exp_{24-29})$	0.005 (0.008)	0.023* (0.013)	0.004 (0.009)	0.016 (0.019)	0.011 (0.011)	0.010 (0.009)
$Avg(Exp_{30+})$	-0.001 (0.008)	-0.012 (0.012)	-0.025*** (0.008)	0.016 (0.019)	-0.005 (0.010)	-0.008 (0.011)
p(all brackets=0)	0.128	0.011	0.000	0.035	0.171	0.152
p(teen effect equal across split)	0.280	0.280	0.145	0.145	0.809	0.809
Observations	86,169	58,768	78,678	22,551	72,641	72,027
Num Localities	567	533	597	529	597	579
Num Countries	37	37	37	37	37	37
Adj R^2	0.182	0.190	0.208	0.157	0.217	0.165
Outcome Mean	0.490	0.428	0.437	0.508	0.440	0.490
Outcome SD	0.500	0.495	0.496	0.500	0.496	0.500

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. Splits on adult characteristics that may themselves respond to formative-year exposure; interpret descriptively. “p(teen effect equal across split)” tests equality of the displayed teenage coefficients in fully interacted models (group-specific slopes, controls, and fixed effects). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.9. Age-Bracket Sweep: Stayers

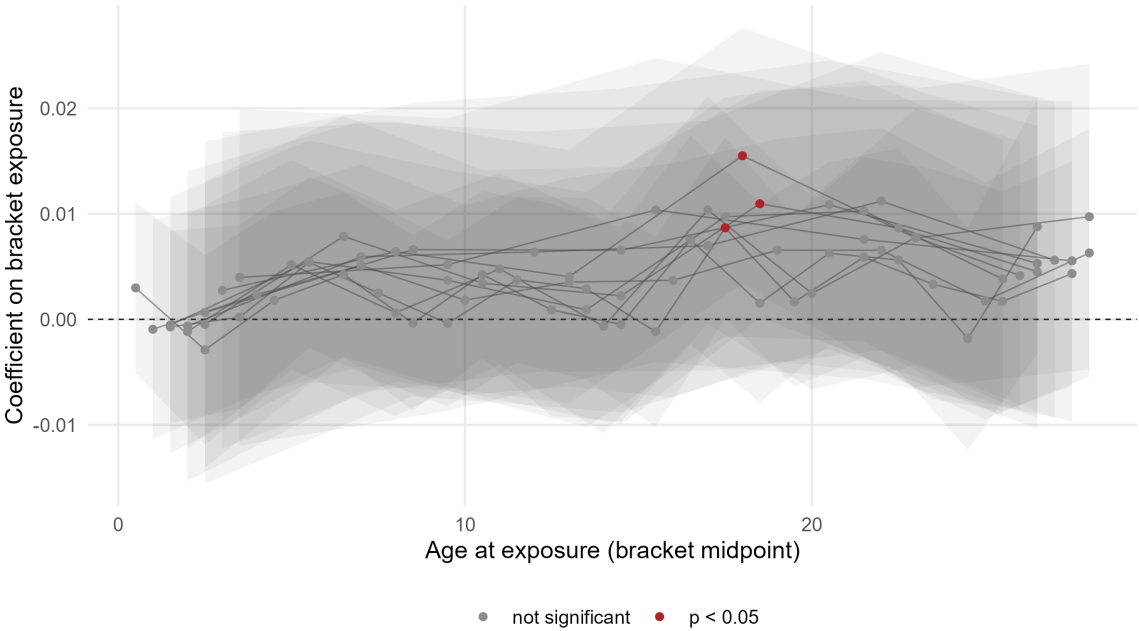
Figure B4: Age-Bracket Sweep: Religious Group Membership, Stayers Only



Notes: As Figure 3, restricting the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. Each line traces the bracket coefficients from one of ten age-bracketing schemes; red points are significant at the 5% level. Of 83 bracket coefficients, none is significant below a bracket midpoint of age 12, 12 of 22 are significant at midpoints 12–19, and 2 of 26 at midpoints in the twenties.

B.10. Age-Bracket Sweep: Community-Group Membership

Figure B5: Age-Bracket Sweep: Drought Exposure and Community-Group Membership



Notes: As Figure 3, with membership in (secular) community associations as the dependent variable. Significant coefficients appear only in brackets containing ages 16–20.

Table B16: Developmental Windows: Community-Group Membership

	Community Member	
	Full	Stayers
	(1)	(2)
$Avg(Exp_{0-11})$	0.010 (0.011)	-0.014 (0.018)
$Avg(Exp_{12-15})$	0.001 (0.006)	-0.001 (0.009)
$Avg(Exp_{16-20})$	0.015** (0.006)	0.011 (0.011)
$Avg(Exp_{21-29})$	0.014 (0.008)	0.011 (0.013)
$Avg(Exp_{30+})$	0.002 (0.008)	0.001 (0.010)
L x Y FE	Y	Y
C x Y x YOB FE	Y	Y
p(all windows=0)	0.099	0.634
p(12-15 = 16-20)	0.100	0.403
Observations	144,723	69,715
Num Localities	599	462
Num Countries	37	30
Adj R^2	0.140	0.132
Outcome Mean	0.362	0.411
Outcome SD	0.481	0.492

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender. “Stayers” restricts the sample to respondents living within 50 km of their matched Murdock ethnic-homeland polygon. *Community Member* refers to membership in a (secular) voluntary community association. Same specification as Table 6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.11. Heterogeneity by Religion and Gender

Religion

Considering that one of the main reasons for understanding the relationship between economic shocks and religiousness in Africa is due to the diverse tapestry of religions present in the country, it becomes impossible to ignore the possibility of heterogeneous effects of drought exposure across different religions. In Appendix B.13, I present coefficient plots for the results from estimating Equation (1) with heterogeneous effects by religion, using the three main religion divisions in the sample: *Christian*, *Muslim*, and *Other* religion — such as Traditionalist or Indigenous. Table B17 in Appendix B.14 reports coefficients and significance tests.

The pattern of effects concentrated during the teenage years is maintained across all three groups: the teenage-window effect on membership is 0.017 ($p < 0.05$) for *Muslims*, 0.019 ($p < 0.05$) for *Christians*, and 0.043 ($p < 0.01$) for adherents of *Other* religions, with no statistically significant differences across groups. The larger point estimate for

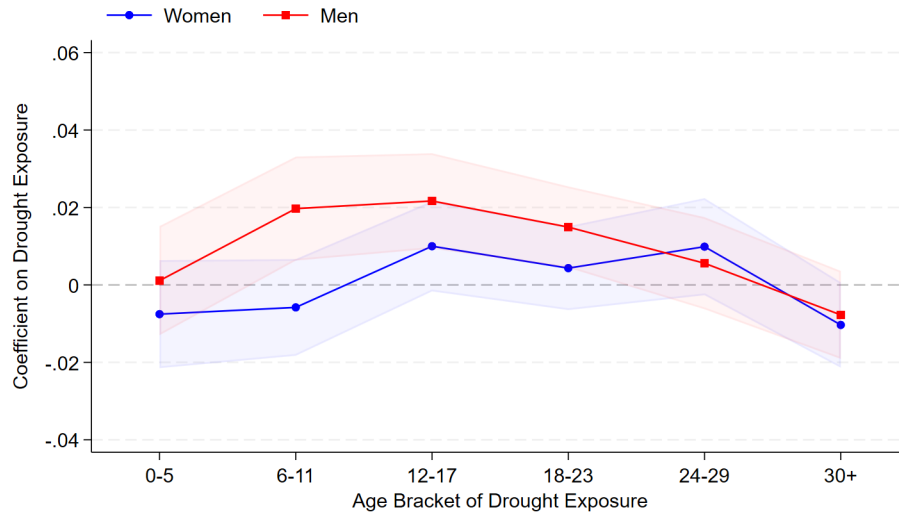
Other religions — largely Traditionalist and Indigenous faiths — extends into neighboring age windows, though the Muslim–Other contrast is at most marginal.

Gender

Additionally, given that religions often have very specific differences in norms or customs across genders, I explore whether there are heterogeneous effects across genders in the coefficient plots reported in Appendix B.12, as well as Table B17 in Appendix B.14. The response is concentrated among men: the teenage-window effect on membership is 0.022 ($p < 0.01$) for men against 0.010 (not significant) for women, a difference that is marginally significant ($p = 0.09$) — this despite men being, on average, around 3 percentage points less likely to be religious and around 10 percentage points less likely to be members of a religious group. The starkest gender difference appears at ages 6–11 ($p < 0.01$ for equality), where men respond (0.020, $p < 0.05$) and women do not. These patterns are likely partially linked to differences in religious practices for men and women at different ages — for example, the performance of male circumcision for Muslim men, *khitan*, is preferred around age 7.

B.12. Droughts and Religiousness: Heterogeneity by Gender

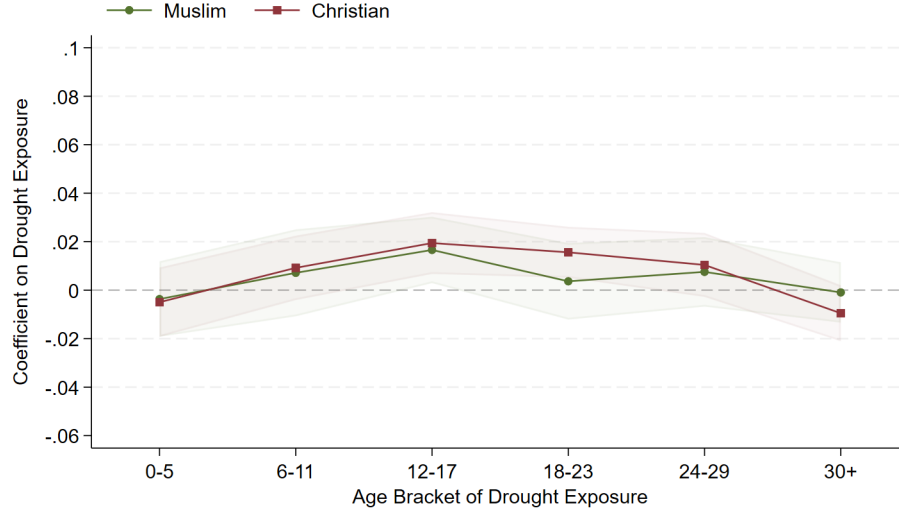
Figure B6: Gender Heterogeneity in Effects of Drought Exposure by Age Bracket



Notes: Estimation of Equation (1) with gender interactions. Shaded area represents 90% confidence intervals.

B.13. Droughts and Religiousness: Heterogeneity by Religion

Figure B7: Religious Heterogeneity in Effects of Drought Exposure by Age Bracket



Notes: Estimation of Equation (1) with Religion interactions. Shaded area represents 90% confidence intervals.

B.14. Droughts and Religiousness: Heterogeneity Summary

Table B17: Impact of Droughts on Religious Group Membership

	Gender			Religion				
	Women (1)	Men (2)	(1) vs (2)	Muslim (3)	Christian (4)	Other (5)	(3) vs (4)	(3) vs (5)
0-5	-0.008 (0.008)	0.001 (0.009)	0.194	-0.004 (0.009)	-0.005 (0.009)	0.033* (0.017)	0.887	0.034
6-11	-0.006 (0.008)	0.020** (0.008)	0.000	0.007 (0.011)	0.009 (0.008)	0.017 (0.017)	0.834	0.545
12-17	0.010 (0.007)	0.022*** (0.007)	0.092	0.017** (0.008)	0.019** (0.008)	0.043*** (0.016)	0.760	0.115
18-23	0.004 (0.007)	0.015** (0.006)	0.058	0.004 (0.010)	0.016** (0.006)	0.012 (0.014)	0.204	0.574
24-29	0.010 (0.008)	0.006 (0.007)	0.317	0.008 (0.009)	0.010 (0.008)	0.015 (0.011)	0.701	0.542
30+	-0.010 (0.007)	-0.008 (0.007)	0.346	-0.001 (0.008)	-0.010 (0.007)	-0.008 (0.009)	0.035	0.298
L x Y FE	Y	Y	Y	Y	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Standard errors clustered by Country-Year in parentheses. L represents Location, Y represents Year, C represents Country, and YOB represents Year of Birth. Controlling for gender. Religious takes a value of 1 if the individual reports being religious and 0 otherwise. Member takes a value of 1 if the individual reports being a member of a religious group and 0 otherwise. Drought exposure is measured at the respondent's current locality, as in the main specification. Restricted to the sample of the main results. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.15. Rites of Passage

Table B18: Does the Effect Run Through Rite-of-Passage Timing?

	Residence exposure		Origin exposure	
	Religious	Member	Religious	Member
$Avg(Exp_{12-17})$	-0.000 (0.004)	0.018** (0.008)		
$Avg(Exp_{12-17}) \times \text{Rite falls in teenage window (ages 1320)}$	-0.005 (0.004)	-0.002 (0.009)		
$Avg(Exp_{12-17,o})$			0.003 (0.003)	-0.005 (0.008)
$Avg(Exp_{12-17,o}) \times \text{Rite falls in teenage window (ages 1320)}$			-0.004 (0.003)	0.010 (0.009)
Rite falls in teenage window (ages 1320)	0.003 (0.006)	0.039*** (0.013)	0.003 (0.005)	0.023* (0.012)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Observations	145,809	145,809	145,809	145,809
Num Localities	599	599	599	599
Num Countries	37	37	37	37
Adj R^2	0.084	0.188	0.084	0.188
Outcome Mean	0.965	0.465	0.965	0.465
Outcome SD	0.183	0.499	0.183	0.499

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents *Year of Birth*. Controlling for gender and for post-rite-window exposure. Each ethnic-religious group is matched to an ethnographic coding of its rite of passage and the rite's age window (compiled from the sources documented in the coding database; high- and medium-confidence codings, or confirmed absence of a rite). "Rite falls in teenage window" indicates that the group's rite window overlaps ages 13–20. The interaction tests whether teenage drought exposure has a larger effect where the formal rite falls in the teenage years; it does not. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.16. Short-Run Cohort Pseudo-Panel

Cells are five-year birth cohorts $\times$ district (GADM-1), built by exact aggregation from single-year-of-birth cells and tracked across Afrobarometer Rounds 5–7 (18,352 cell-year rows). All regressions absorb cell, year, and country-round fixed effects, control for the cell's gender and religious composition, weight by cell size, and cluster standard errors at the district level, where the drought treatment varies. Gender and religion are not used to define cells: religion is a switchable characteristic, so cells defined on it would not contain the same individuals across rounds.

Table B19: Short-Run Effects of Severe Drought: Within-Cohort Estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	Member	Member	Member	Food shortage	Food shortage	Employed
Severe drought months, $t-1$	0.008*** (0.003)	0.008** (0.003)	0.008*** (0.003)		-0.002 (0.003)	0.001 (0.003)
Ages 18-27 now=1		-0.011 (0.010)				
Ages 18-27 now=1 $\times$ Severe drought months, $t-1$		0.001 (0.002)				
Severe drought months, $t-1 \times$ Current age (decades from 40)			0.000 (0.001)			
Severe drought months, t				0.004 (0.003)		
Constant	0.066* (0.038)	0.069* (0.038)	0.066* (0.038)	0.597*** (0.058)	0.613*** (0.035)	0.278*** (0.039)
Observations	14204	14204	14204	10909	14218	14208
Adjusted R^2	0.577	0.577	0.576	0.573	0.570	0.562
Mean Dep. Var.	0.479	0.479	0.479	0.489	0.485	0.364
Effect for 18-27 (lincom)		0.009				
... se		0.003				
... p-value		0.008				

Notes: Birth-cohort pseudo-panel. "Severe drought months" counts months of severe drought in the district over the named year (t is the survey year). "Ages 18–27 now" refers to the cohort's current age at the survey; its main effect is identified because cohorts age across rounds within a cell. The food-shortage outcome is the cell share reporting having gone without food in the past year, so contemporaneous drought is the relevant timing; the $t - 1$ column is shown for comparison. Standard errors clustered by district. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B20: Cohort-Band Robustness: Severe-Drought Months

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	3/0	3/1	4/0	4/2	5/0	5/2	10/0	10/5
Severe drought months, $t-1$	0.007** (0.003)	0.008** (0.003)	0.008** (0.003)	0.008** (0.003)	0.008*** (0.003)	0.007** (0.003)	0.008*** (0.003)	0.008** (0.003)
Observations	20573	20612	16960	17069	14204	14447	8710	8112
Adjusted R^2	0.511	0.503	0.546	0.543	0.577	0.575	0.659	0.663

Notes: Each column rebuilds the pseudo-panel under a different cohort-band scheme, labeled width/offset in years, and re-estimates the pooled specification of column (1) of Table B19. The corresponding estimates for the annual-minimum measure are between 0.005 and 0.009 and insignificant in every scheme. Standard errors clustered by district. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B21: Short-Run Effects, Annual-Minimum Measure: Within-Cohort Estimates

	(1) Member	(2) Member	(3) Member	(4) Member	(5) Employed
Drought (min SPEI), t-1	0.006 (0.010)	0.008 (0.010)			-0.015 (0.012)
Ages 18-27 now=1		-0.004 (0.012)		-0.007 (0.020)	
Ages 18-27 now=1 × Drought (min SPEI), t-1		-0.004 (0.007)			
Drought (min SPEI), 3-yr avg			-0.020 (0.033)	-0.015 (0.033)	
Ages 18-27 now=1 × Drought (min SPEI), 3-yr avg				-0.015 (0.013)	
Constant	0.067* (0.040)	0.068* (0.040)	0.144** (0.067)	0.145** (0.068)	0.296*** (0.042)
Observations	14204	14204	10900	10900	14208
Adjusted R^2	0.575	0.575	0.619	0.619	0.562
Mean Dep. Var.	0.479	0.479	0.468	0.468	0.364
Effect for 18-27 (lincom)		0.003		-0.030	
... se		0.011		0.034	
... p-value		0.764		0.373	

Notes: As Table B19, with the annual-minimum drought measure used in the main analysis and its three-year average.
Standard errors clustered by district. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B22: Short-Run Effects by Gender: Within-Cohort Estimates

	(1) Women	(2) Men	(3) Pooled
Severe drought months, t-1	0.010*** (0.003)	0.006* (0.003)	0.008** (0.003)
Male=1 × Severe drought months, t-1			0.000 (0.002)
Constant	0.016 (0.048)	0.108*** (0.031)	0.075*** (0.028)
Observations	10930	12147	23077
Adjusted R^2	0.536	0.440	0.489
Mean Dep. Var.	0.483	0.470	0.476
p(effect equal by gender)			0.815

Notes: Cells defined by five-year birth cohort × district × gender (gender is fixed at birth, so gender-specific cells preserve cohort identity). Standard errors clustered by district. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.17. Mechanisms and other Outcomes

B.17.1. Economic Conditions Results

Table B23: Impact of Droughts on Lack of Access to Resources

	Dependent Variable:				
	Food	Water	Medical	Fuel	Cash
	(1)	(2)	(3)	(4)	(5)
<i>Avg(Exp₀₋₅)</i>	0.009 (0.007)	-0.004 (0.006)	0.003 (0.007)	0.008 (0.008)	-0.002 (0.005)
<i>Avg(Exp₆₋₁₁)</i>	-0.000 (0.007)	-0.002 (0.008)	-0.003 (0.007)	-0.005 (0.007)	-0.004 (0.006)
<i>Avg(Exp₁₂₋₁₇)</i>	0.003 (0.007)	0.005 (0.007)	0.021*** (0.008)	0.011 (0.008)	0.006 (0.007)
<i>Avg(Exp₁₈₋₂₃)</i>	0.012* (0.006)	0.007 (0.008)	0.003 (0.007)	0.001 (0.006)	0.000 (0.005)
<i>Avg(Exp₂₄₋₂₉)</i>	0.008 (0.006)	-0.003 (0.006)	0.010 (0.007)	-0.002 (0.006)	0.010* (0.006)
<i>Avg(Exp₃₀₊)</i>	-0.001 (0.007)	0.000 (0.007)	0.007 (0.008)	0.010 (0.007)	-0.004 (0.008)
L x Y FE	Y	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y	Y
Observations	145,727	145,729	145,391	145,392	145,425
Num Localities	599	599	599	599	599
Num Countries	37	37	37	37	37
Adj <i>R</i> ²	0.169	0.142	0.159	0.135	0.234
Outcome Mean	0.524	0.523	0.481	0.613	0.236
Outcome SD	0.499	0.499	0.500	0.487	0.424

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. Controlling for gender. *Religious* takes a value of 1 if the individual reports being religious and 0 otherwise. *Member* takes a value of 1 if the individual reports being a member of a religious group and 0 otherwise. Drought exposure is measured at the respondent's current locality, as in the main specification. Restricted to the sample of the main results. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.17.2. Community Ties

Table B24: Impact of Droughts on Community Ties

	Dependent Variable:			
	Community Group Member	Trust Electoral Commission	Trust Opposition Parties	Trust Police
	(1)	(2)	(3)	(4)
<i>Avg(Exp₀₋₅)</i>	0.001 (0.008)	0.001 (0.007)	-0.004 (0.007)	0.006 (0.007)
<i>Avg(Exp₆₋₁₁)</i>	0.007 (0.007)	-0.015** (0.007)	-0.011 (0.007)	0.016** (0.007)
<i>Avg(Exp₁₂₋₁₇)</i>	0.007 (0.008)	0.005 (0.008)	0.006 (0.006)	0.004 (0.007)
<i>Avg(Exp₁₈₋₂₃)</i>	0.011 (0.007)	0.007 (0.007)	0.003 (0.006)	0.015** (0.007)
<i>Avg(Exp₂₄₋₂₉)</i>	0.005 (0.008)	-0.007 (0.008)	-0.024*** (0.006)	0.009 (0.007)
<i>Avg(Exp₃₀₊)</i>	-0.001 (0.007)	-0.007 (0.008)	-0.011 (0.008)	0.017** (0.008)
L x Y FE	Y	Y	Y	Y
C x Y x YOB FE	Y	Y	Y	Y
Observations	144,723	131,691	133,172	143,127
Num Localities	599	599	595	599
Num Countries	37	37	36	37
Adj <i>R</i> ²	0.140	0.140	0.082	0.135
Outcome Mean	0.362	0.516	0.375	0.516
Outcome SD	0.481	0.500	0.484	0.500

Notes: Standard errors clustered by Country-Year in parentheses. L represents *Location*, Y represents *Year*, C represents *Country*, and YOB represents Year of Birth. Controlling for gender. *Community Group Member* takes a value of 1 if the individual reports being a member of a (secular) voluntary community association; the trust outcomes are indicators for trusting the named institution "somewhat" or "a lot". Drought exposure is measured at the respondent's current locality, as in the main specification. Restricted to the sample of the main results. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.